Chapter 7

Maximum Likelihood Estimation

7.1 Introduction

We now investigate an alternative to the MVU estimator, which is desirable in situations where the MVU estimator does not exist or cannot be found even if it does exist. This estimator, which is based on the maximum likelihood principle, is overwhelmingly the most popular approach to obtaining *practical* estimators. It has the distinct advantage of being a "turn-the-crank" procedure, allowing it to be implemented for complicated estimation problems. Additionally, for most cases of practical interest its performance is optimal for large enough data records. Specifically, it is approximately the MVU estimator due to its approximate efficiency. For these reasons almost all practical estimators are based on the maximum likelihood principle.

7.2 Summary

In Section 7.3 we discuss an estimation problem for which the CRLB cannot be achieved and hence an efficient estimator does not exist, and for which the sufficient statistic approach in Chapter 5 to finding the MVU estimator cannot be implemented. An estimator is examined, however, that is asymptotically (for large data records) efficient. This is the maximum likelihood estimator (MLE), defined as the value of θ that maximizes the likelihood function. In general, as stated in Theorem 7.1, the MLE has the asymptotic properties of being unbiased, achieving the CRLB, and having a Gaussian PDF. Thus, it can be said to be asymptotically optimal. In Example 7.6 it is also shown that for signal in noise problems, the MLE achieves the CRLB for high SNRs. The MLE of θ can be used to find the MLE of a function of θ by invoking the invariance property as described by Theorem 7.2. When a closed form expression cannot be found for the MLE, a numerical approach employs either a grid search or an iterative maximization of the likelihood function. The iterative approaches, which are used only when a grid search is not practical, are the Newton-Raphson (7.29) and scoring (7.33) methods. Convergence to the MLE, however, is not guaranteed. The MLE for a vector parameter $\boldsymbol{\theta}$ is the value maximizing the likelihood function, which is now a

function of the components of $\boldsymbol{\theta}$. The asymptotic properties of the vector parameter MLE are summarized in Theorem 7.3, and the invariance property in Theorem 7.4. For the linear model the MLE achieves the CRLB for finite data records. The resultant estimator is thus the usual MVU estimator as summarized in Theorem 7.5. Numerical determination of the vector parameter MLE is based on the Newton-Raphson (7.48) or scoring (7.50) approach. Another technique is the expectation-maximization (EM) algorithm of (7.56) and (7.57), which is iterative in nature. Finally, for WSS Gaussian random processes the MLE can be found approximately by maximizing the asymptotic log-likelihood function of (7.60). This approach is computationally less intensive than determination of the exact MLE.

7.3 An Example

As a rationale for why we might be interested in an approximately optimal estimator, we now examine an estimation problem for which there is no obvious way to find the MVU estimator. By resorting to the estimator based on the maximum likelihood principle, termed the *maximum likelihood estimator* (MLE), we can obtain an estimator that is approximately the MVU estimator. The nature of the approximation relies on the property that the MLE is asymptotically (for large data records) efficient.

Example 7.1 - DC Level in White Gaussian Noise - Modified

Consider the observed data set

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where A is an unknown level, which is assumed to be positive ($A > 0$), and $w[n]$ is WGN *with unknown variance* A. This problem differs from our usual one (see Example 3.3) in that the unknown parameter A is reflected in the mean *and* the variance. In searching for the MVU estimator of A we first determine the CRLB (see Chapter 3) to see if it is satisfied with equality. The PDF is

$$p(\mathbf{x}; A) = \frac{1}{(2\pi A)^{\frac{N}{2}}} \exp\left[-\frac{1}{2A}\sum_{n=0}^{N-1}(x[n]-A)^2\right]. \tag{7.1}$$

Taking the derivative of the log-likelihood function (the logarithm of the PDF considered as a function of A), we have

$$\begin{aligned}\frac{\partial \ln p(\mathbf{x}; A)}{\partial A} &= -\frac{N}{2A} + \frac{1}{A}\sum_{n=0}^{N-1}(x[n]-A) + \frac{1}{2A^2}\sum_{n=0}^{N-1}(x[n]-A)^2 \\ &\stackrel{?}{=} I(A)(\hat{A}-A).\end{aligned}$$

It is certainly not obvious if the derivative of the log-likelihood function can be put in the required form. It appears from a casual observation that it cannot, and therefore,

an efficient estimator does not exist. We can still determine the CRLB for this problem to find that (see Problem 7.1)

$$\text{var}(\hat{A}) \geq \frac{A^2}{N(A+\frac{1}{2})}. \tag{7.2}$$

We next try to find the MVU estimator by resorting to the theory of sufficient statistics (see Chapter 5). Attempting to factor (7.1) into the form of (5.3), we note that

$$\frac{1}{A}\sum_{n=0}^{N-1}(x[n]-A)^2 = \frac{1}{A}\sum_{n=0}^{N-1}x^2[n] - 2N\bar{x} + NA$$

so that the PDF factors as

$$p(\mathbf{x};A) = \underbrace{\frac{1}{(2\pi A)^{\frac{N}{2}}}\exp\left[-\frac{1}{2}\left(\frac{1}{A}\sum_{n=0}^{N-1}x^2[n]+NA\right)\right]}_{g\left(\sum_{n=0}^{N-1}x^2[n],A\right)}\underbrace{\exp(N\bar{x})}_{h(\mathbf{x})}.$$

Based on the Neyman-Fisher factorization theorem a single sufficient statistic for A is $T(\mathbf{x}) = \sum_{n=0}^{N-1}x^2[n]$. The next step is to find a function of the sufficient statistic that produces an unbiased estimator, assuming that $T(\mathbf{x})$ is a *complete* sufficient statistic. To do so we need to find a function g such that

$$E\left[g\left(\sum_{n=0}^{N-1}x^2[n]\right)\right] = A \qquad \text{for all } A>0.$$

Since

$$\begin{aligned} E\left[\sum_{n=0}^{N-1}x^2[n]\right] &= NE[x^2[n]] \\ &= N\left[\text{var}(x[n]) + E^2(x[n])\right] \\ &= N(A+A^2) \end{aligned}$$

it is not obvious how to choose g. We cannot simply scale the sufficient statistic to make it unbiased as we did in Example 5.8.

A second approach would be to determine the conditional expectation $E(\hat{A}|\sum_{n=0}^{N-1}x^2[n])$, where $\hat{A}$ is any unbiased estimator. As an example, if we were to choose the unbiased estimator $\hat{A} = x[0]$, then the MVU estimator would take the form

$$E\left(x[0]\middle|\sum_{n=0}^{N-1}x^2[n]\right). \tag{7.3}$$

Unfortunately, the evaluation of the conditional expectation appears to be a formidable task.

We have now exhausted our possible optimal approaches. This is not to say that we could not propose some estimators. One possibility considers A to be the mean, so that

$$\hat{A}_1 = \begin{cases} \bar{x} & \text{if } \bar{x} > 0 \\ 0 & \text{if } \bar{x} \leq 0 \end{cases}$$

since we know that $A > 0$. Another estimator considers A to be the variance, to yield

$$\hat{A}_2 = \frac{1}{N-1} \sum_{n=0}^{N-1} (x[n] - \hat{A}_1)^2$$

(see also Problem 7.2). However, these estimators cannot be claimed to be optimal in any sense. ◇

Faced with our inability to find the MVU estimator we will propose an estimator that is *approximately optimal*. We claim that for large data records or as $N \to \infty$, the proposed estimator is efficient. This means that as $N \to \infty$

$$E(\hat{A}) \to A \tag{7.4}$$

$$\text{var}(\hat{A}) \to \text{CRLB}, \tag{7.5}$$

the CRLB being given by (7.2). An estimator $\hat{A}$ that satisfies (7.4) is said to be *asymptotically unbiased*. If, in addition, the estimator satisfies (7.5), then it is said to be *asymptotically efficient*. For finite data records, however, we can say nothing about its optimality (see Problem 7.4). Better estimators may exist, but finding them may not be easy!

Example 7.2 - DC Level in White Gaussian Noise - Modified (continued)

We propose the following estimator:

$$\hat{A} = -\frac{1}{2} + \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} x^2[n] + \frac{1}{4}}. \tag{7.6}$$

This estimator is biased since

$$\begin{aligned} E(\hat{A}) &= E\left(-\frac{1}{2} + \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} x^2[n] + \frac{1}{4}}\right) \\ &\neq -\frac{1}{2} + \sqrt{E\left(\frac{1}{N} \sum_{n=0}^{N-1} x^2[n]\right) + \frac{1}{4}} \qquad \text{for all } A \\ &= -\frac{1}{2} + \sqrt{A + A^2 + \frac{1}{4}} \\ &= A. \end{aligned}$$

It is nonetheless a reasonable estimator in that as $N \to \infty$, we have by the law of large numbers

$$\frac{1}{N}\sum_{n=0}^{N-1} x^2[n] \to E(x^2[n]) = A + A^2$$

and therefore from (7.6)

$$\hat{A} \to A.$$

The estimator $\hat{A}$ is said to be a *consistent* estimator (see Problems 7.5 and 7.6). To find the mean and variance of $\hat{A}$ as $N \to \infty$ we use the statistical linearization argument described in Section 3.6. For this example the PDF of $\frac{1}{N}\sum_{n=0}^{N-1} x^2[n]$ will be concentrated about its mean, $A + A^2$, for large data records. This allows us to linearize the function given in (7.6) that transforms $\frac{1}{N}\sum_{n=0}^{N-1} x^2[n]$ into $\hat{A}$. Let g be that function, so that

$$\hat{A} = g(u)$$

where $u = \frac{1}{N}\sum_{n=0}^{N-1} x^2[n]$, and therefore,

$$g(u) = -\frac{1}{2} + \sqrt{u + \frac{1}{4}}.$$

Linearizing about $u_0 = E(u) = A + A^2$, we have

$$g(u) \approx g(u_0) + \left.\frac{dg(u)}{du}\right|_{u=u_0} (u - u_0)$$

or

$$\hat{A} \approx A + \frac{\frac{1}{2}}{A + \frac{1}{2}} \left[\frac{1}{N}\sum_{n=0}^{N-1} x^2[n] - (A + A^2)\right]. \tag{7.7}$$

It now follows that the asymptotic mean is

$$E(\hat{A}) = A$$

so that $\hat{A}$ is asymptotically unbiased. Additionally, the asymptotic variance becomes, from (7.7),

$$\begin{aligned} \text{var}(\hat{A}) &= \left(\frac{\frac{1}{2}}{A + \frac{1}{2}}\right)^2 \text{var}\left[\frac{1}{N}\sum_{n=0}^{N-1} x^2[n]\right] \\ &= \frac{\frac{1}{4}}{N(A + \frac{1}{2})^2} \text{var}(x^2[n]). \end{aligned}$$

But $\text{var}(x^2[n])$ can be shown to be $4A^3 + 2A^2$ (see Section 3.6), so that

$$\begin{aligned} \text{var}(\hat{A}) &= \frac{\frac{1}{4}}{N(A + \frac{1}{2})^2} 4A^2(A + \frac{1}{2}) \\ &= \frac{A^2}{N(A + \frac{1}{2})} \end{aligned}$$

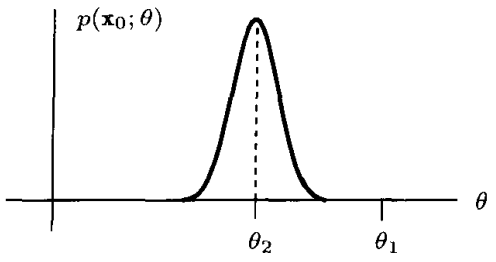


Figure 7.1 Rationale for maximum likelihood estimator

which from (7.2) is the CRLB!

Summarizing our results, the proposed estimator given by (7.6) is asymptotically unbiased and asymptotically achieves the CRLB. Hence, it is *asymptotically efficient.* Furthermore, by the central limit theorem the random variable $\frac{1}{N}\sum_{n=0}^{N-1} x^2[n]$ is Gaussian as $N \to \infty$. Because $\hat{A}$ is a linear function of this Gaussian random variable for large data records (as per (7.7)), it too will have a Gaussian PDF. $\diamond$

The proposed estimator is termed the MLE. How it was found is discussed in the next section.

7.4 Finding the MLE

The MLE for a scalar parameter is defined to be *the value of θ that maximizes $p(\mathbf{x};\theta)$* for $\mathbf{x}$ fixed, i.e., the value that maximizes the likelihood function. The maximization is performed over the allowable range of θ. In the previous example this was $A > 0$. Since $p(\mathbf{x};\theta)$ will also be a function of $\mathbf{x}$, the maximization produces a $\hat{\theta}$ that is a function of $\mathbf{x}$. The rationale for the MLE hinges on the observation that $p(\mathbf{x};\theta)\,d\mathbf{x}$ gives the probability of observing $\mathbf{x}$ in a small volume for a given θ. In Figure 7.1 the PDF is evaluated for $\mathbf{x} = \mathbf{x}_0$ and then plotted versus θ. The value of $p(\mathbf{x} = \mathbf{x}_0;\theta)\,d\mathbf{x}$ for each θ tells us the probability of observing $\mathbf{x}$ in the region in R^N centered around $\mathbf{x}_0$ with volume $d\mathbf{x}$, assuming the given value of θ. If $\mathbf{x} = \mathbf{x}_0$ had indeed been observed, then inferring that $\theta = \theta_1$ would be unreasonable. Because if $\theta = \theta_1$, the probability of actually observing $\mathbf{x} = \mathbf{x}_0$ would be small. It is more "likely" that $\theta = \theta_2$ is the true value. It yields a high probability of observing $\mathbf{x} = \mathbf{x}_0$, the data that were *actually observed.* Thus, we choose $\hat{\theta} = \theta_2$ as our estimate or the value that maximizes $p(\mathbf{x} = \mathbf{x}_0;\theta)$ over the allowable range of θ. We now continue with our example.

Example 7.3 - DC Level in White Gaussian Noise - Modified (continued)

To actually find the MLE for this problem we first write the PDF from (7.1) as

$$p(\mathbf{x};A) = \frac{1}{(2\pi A)^{\frac{N}{2}}} \exp\left[-\frac{1}{2A}\sum_{n=0}^{N-1}(x[n]-A)^2\right].$$

Considering this as a function of A, it becomes the likelihood function. Differentiating the log-likelihood function, we have

$$\frac{\partial \ln p(\mathbf{x}; A)}{\partial A} = -\frac{N}{2A} + \frac{1}{A}\sum_{n=0}^{N-1}(x[n] - A) + \frac{1}{2A^2}\sum_{n=0}^{N-1}(x[n] - A)^2$$

and setting it equal to zero produces

$$\hat{A}^2 + \hat{A} - \frac{1}{N}\sum_{n=0}^{N-1} x^2[n] = 0.$$

Solving for $\hat{A}$ produces the two solutions

$$\hat{A} = -\frac{1}{2} \pm \sqrt{\frac{1}{N}\sum_{n=0}^{N-1} x^2[n] + \frac{1}{4}}.$$

We choose the solution

$$\hat{A} = -\frac{1}{2} + \sqrt{\frac{1}{N}\sum_{n=0}^{N-1} x^2[n] + \frac{1}{4}}$$

to correspond to the permissible range of A or $A > 0$. Note that $\hat{A} > 0$ for all possible values of $\frac{1}{N}\sum_{n=0}^{N-1} x^2[n]$. Finally, that $\hat{A}$ indeed maximizes the log-likelihood function is verified by examining the second derivative. ◇

Not only does the maximum likelihood procedure yield an estimator that is asymptotically efficient, it also sometimes yields an efficient estimator for *finite* data records. This is illustrated in the following example.

Example 7.4 - DC Level in White Gaussian Noise

For the received data

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where A is the unknown level to be estimated and $w[n]$ is WGN with known variance σ^2, the PDF is

$$p(\mathbf{x}; A) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n] - A)^2\right].$$

Taking the derivative of the log-likelihood function produces

$$\frac{\partial \ln p(\mathbf{x}; A)}{\partial A} = \frac{1}{\sigma^2}\sum_{n=0}^{N-1}(x[n] - A)$$

which being set equal to zero yields the MLE

$$\hat{A} = \frac{1}{N}\sum_{n=0}^{N-1} x[n].$$

But we have already seen that the sample mean is an efficient estimator (see Example 3.3). Hence, the MLE is efficient. ◇

This result is true in general. *If an efficient estimator exists, the maximum likelihood procedure will produce it.* See Problem 7.12 for an outline of the proof.

7.5 Properties of the MLE

The example discussed in Section 7.3 led to an estimator that for large data records (or asymptotically) was unbiased, achieved the CRLB, and had a Gaussian PDF. In summary, the MLE was distributed as

$$\hat{\theta} \overset{a}{\sim} \mathcal{N}(\theta, I^{-1}(\theta)) \tag{7.8}$$

where $\overset{a}{\sim}$ denotes "asymptotically distributed according to." This result is quite general and forms the basis for claiming optimality of the MLE. Of course, in practice it is seldom known in advance how large N must be in order for (7.8) to hold. An analytical expression for the PDF of the MLE is usually impossible to derive. As an alternative means of assessing performance, a computer simulation is usually required, as discussed next.

Example 7.5 - DC Level in White Gaussian Noise - Modified (continued)

A computer simulation was performed to determine how large the data record had to be for the asymptotic results to apply. In principle the exact PDF of $\hat{A}$ (see (7.6)) could be found but would be extremely tedious. Using the Monte Carlo method (see Appendix 7A), $M = 1000$ realizations of $\hat{A}$ were generated for various data record lengths. The mean $E(\hat{A})$ and variance $\text{var}(\hat{A})$ were *estimated* by

$$\widehat{E(\hat{A})} = \frac{1}{M}\sum_{i=1}^{M} \hat{A}_i \tag{7.9}$$

$$\widehat{\text{var}(\hat{A})} = \frac{1}{M}\sum_{i=1}^{M} \left(\hat{A}_i - \widehat{E(\hat{A})}\right)^2. \tag{7.10}$$

For a value of A equal to 1 the results are shown in Table 7.1 for various data record lengths. Instead of the asymptotic variance or the CRLB of (7.2), we tabulate

$$N\text{var}(\hat{A}) = \frac{A^2}{A + \frac{1}{2}}$$

TABLE 7.1 Theoretical Asymptotic and Actual Mean and Variance for Estimator in Example 7.2

Data Record Length, N	Mean, $E(\hat{A})$	$N \times$Variance, $N\,\mathrm{var}(\hat{A})$
5	0.954	0.624
10	0.976	0.648
15	0.991	0.696
20	0.996 (0.987)	0.707 (0.669)
25	0.994	0.656
Theoretical asymptotic value	1	0.667

since this is independent of N, allowing us to check the convergence more readily. The theoretical asymptotic values of the mean and normalized variance are

$$\begin{aligned} E(\hat{A}) &= A = 1 \\ N\mathrm{var}(\hat{A}) &= \frac{2}{3}. \end{aligned}$$

It is observed from Table 7.1 that the mean converges at about $N = 20$ samples, while the variance jumps around somewhat for $N \geq 15$ samples. The latter is due to the statistical fluctutations in estimating the variance via a computer simulation, as well as possible inaccuracies in the random number generator. To check this the number of realizations was increased to $M = 5000$ for a data record length of $N = 20$. This resulted in the mean and normalized variance shown in parentheses. The normalized variance is now nearly identical to its asymptotic value, whereas for some unknown reason (presumably the random number generator) the mean is off slightly from its asymptotic value. (See also Problem 9.8 for a more accurate formula for $E(\hat{A})$.)

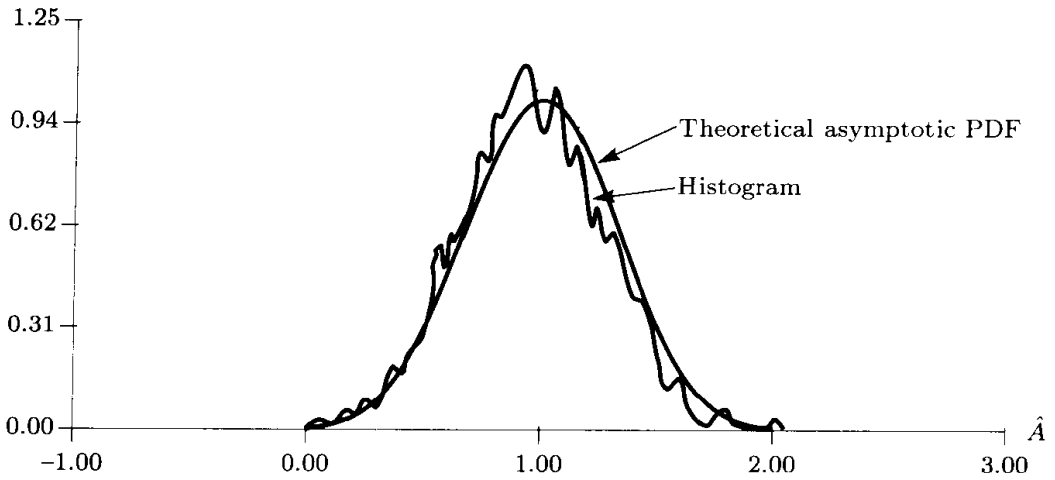
Next, the PDF of $\hat{A}$ was determined using a Monte Carlo computer simulation. This was done for data record lengths of $N = 5$ and $N = 20$. According to (7.8), the asymptotic PDF is

$$\hat{A} \overset{a}{\sim} \mathcal{N}(A, \boldsymbol{I}^{-1}(A)),$$

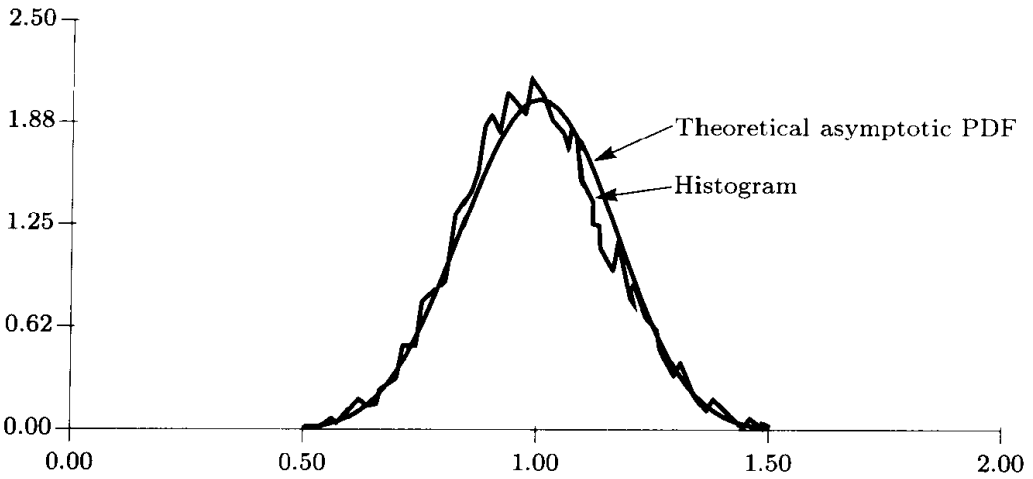
which for $A = 1$ becomes, upon using (7.2),

$$\hat{A} \overset{a}{\sim} \mathcal{N}(1, \frac{2/3}{N}).$$

In Figure 7.2 the theoretical PDF and estimated PDF or *histogram* (see Appendix 7A) are shown. To construct the histogram we used $M = 5000$ realizations of $\hat{A}$ and divided the horizontal axis into 100 cells or divisions. Note that for $N = 5$ the estimated PDF is somewhat displaced to the left, in accordance with the mean being too small (see Table 7.1). For $N = 20$, however, the match is better, although the estimated PDF still appears to be skewed to the left. Presumably for larger data records the asymptotic PDF will more closely match the true one. ◇



(a) $N = 5$



(b) $N = 20$

Figure 7.2 Theoretical PDF and histogram

In describing the performance of estimators whose PDFs cannot be determined analytically, we must frequently resort to computer simulations, as in the previous example. In practice, the computer has become the dominant tool for analyzing the performance of nonlinear estimators. For this reason it is important to be able to carry out such a simulation. In Appendix 7A a description of the computer methods used in the previous example is given. Of course, the subject of Monte Carlo computer methods for statistical evaluation warrants a more complete discussion. The interested reader should consult the following references: [Bendat and Piersol 1971, Schwartz and Shaw

1975]. Some practice in performing these simulations is provided by Problems 7.13 and 7.14. We now summarize the asymptotic properties of the MLE in a theorem.

Theorem 7.1 (Asymptotic Properties of the MLE) *If the PDF $p(\mathbf{x};\theta)$ of the data $\mathbf{x}$ satisfies some "regularity" conditions, then the MLE of the unknown parameter θ is asymptotically distributed (for large data records) according to*

$$\hat{\theta} \overset{a}{\sim} \mathcal{N}(\theta, I^{-1}(\theta)) \tag{7.11}$$

where $I(\theta)$ is the Fisher information evaluated at the true value of the unknown parameter.

The regularity conditions require the existence of the derivatives of the log-likelihood function, as well as the Fisher information being nonzero, as described more fully in Appendix 7B. An outline of the proof for IID observations is also given there.

From the asymptotic distribution, the MLE is seen to be asymptotically unbiased and asymptotically attains the CRLB. It is therefore *asymptotically efficient*, and hence *asymptotically optimal*. Of course, in practice the key question is always How large does N have to be for the asymptotic properties to apply? Fortunately, for many cases of interest the data record lengths are not excessive, as illustrated by Example 7.5. Another example follows.

Example 7.6 - MLE of the Sinusoidal Phase

We now reconsider the problem in Example 3.4 in which we wish to estimate the phase ϕ of a sinusoid embedded in noise or

$$x[n] = A\cos(2\pi f_0 n + \phi) + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN with variance σ^2 and the amplitude A and frequency f_0 are assumed to be known. We saw in Chapter 5 that no *single* sufficient statistic exists for this problem. The sufficient statistics were

$$\begin{aligned} T_1(\mathbf{x}) &= \sum_{n=0}^{N-1} x[n]\cos(2\pi f_0 n) \\ T_2(\mathbf{x}) &= \sum_{n=0}^{N-1} x[n]\sin(2\pi f_0 n). \end{aligned} \tag{7.12}$$

The MLE is found by maximizing $p(\mathbf{x};\phi)$ or

$$p(\mathbf{x};\phi) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}\left(x[n] - A\cos(2\pi f_0 n + \phi)\right)^2\right]$$

or, equivalently, by minimizing

$$J(\phi) = \sum_{n=0}^{N-1}\left(x[n] - A\cos(2\pi f_0 n + \phi)\right)^2. \tag{7.13}$$

Differentiating with respect to ϕ produces

$$\frac{\partial J(\phi)}{\partial \phi} = 2 \sum_{n=0}^{N-1} (x[n] - A\cos(2\pi f_0 n + \phi))\, A \sin(2\pi f_0 n + \phi)$$

and setting it equal to zero yields

$$\sum_{n=0}^{N-1} x[n] \sin(2\pi f_0 n + \hat{\phi}) = A \sum_{n=0}^{N-1} \sin(2\pi f_0 n + \hat{\phi}) \cos(2\pi f_0 n + \hat{\phi}). \tag{7.14}$$

But the right-hand side may be approximated since (see Problem 3.7)

$$\frac{1}{N} \sum_{n=0}^{N-1} \sin(2\pi f_0 n + \hat{\phi}) \cos(2\pi f_0 n + \hat{\phi}) = \frac{1}{2N} \sum_{n=0}^{N-1} \sin(4\pi f_0 n + 2\hat{\phi}) \approx 0 \tag{7.15}$$

for f_0 not near 0 or 1/2. Thus, the left-hand side of (7.14) when divided by N and set equal to zero will produce an approximate MLE, which satisfies

$$\sum_{n=0}^{N-1} x[n] \sin(2\pi f_0 n + \hat{\phi}) = 0. \tag{7.16}$$

Upon expanding this we have

$$\sum_{n=0}^{N-1} x[n] \sin 2\pi f_0 n \cos \hat{\phi} = - \sum_{n=0}^{N-1} x[n] \cos 2\pi f_0 n \sin \hat{\phi}$$

or finally the MLE of phase is given approximately as

$$\hat{\phi} = - \arctan \frac{\sum_{n=0}^{N-1} x[n] \sin 2\pi f_0 n}{\sum_{n=0}^{N-1} x[n] \cos 2\pi f_0 n}. \tag{7.17}$$

It is interesting to note that the MLE is a function of the sufficient statistics. In hindsight, this should not be surprising if we keep in mind the Neyman-Fisher factorization theorem. In this example there are two sufficient statistics, effecting a factorization as

$$p(\mathbf{x}; \phi) = g(T_1(\mathbf{x}), T_2(\mathbf{x}), \phi) h(\mathbf{x}).$$

Clearly, maximizing $p(\mathbf{x}; \phi)$ is equivalent to maximizing g ($h(\mathbf{x})$ can always be chosen so that $h(\mathbf{x}) > 0$), and thus $\hat{\phi}$ must be a function of $T_1(\mathbf{x})$ and $T_2(\mathbf{x})$.

According to Theorem 7.1, the asymptotic PDF of the phase estimator is

$$\hat{\phi} \overset{a}{\sim} \mathcal{N}(\phi, I^{-1}(\phi)). \tag{7.18}$$

TABLE 7.2 Theoretical Asymptotic and Actual Mean and Variance for Estimator in Example 7.6

Data Record Length, N	Mean, $E(\hat{\phi})$	$N \times$Variance, $N\,\mathrm{var}(\hat{\phi})$
20	0.732	0.0978
40	0.746	0.108
60	0.774	0.110
80	0.789	0.0990
Theoretical asymptotic value	$\phi = 0.785$	$1/\eta = 0.1$

From Example 3.4

$$I(\phi) = \frac{NA^2}{2\sigma^2}$$

so that the asymptotic variance is

$$\mathrm{var}(\hat{\phi}) = \frac{1}{N\frac{A^2}{2\sigma^2}} = \frac{1}{N\eta} \tag{7.19}$$

where $\eta = (A^2/2)/\sigma^2$ is the SNR. To determine the data record length for the asymptotic mean and variance to apply we performed a computer simulation using $A = 1, f_0 = 0.08, \phi = \pi/4$, and $\sigma^2 = 0.05$. The results are listed in Table 7.2. It is seen that the asymptotic mean and normalized variance are attained for $N = 80$. For shorter data records the estimator is considerably biased. Part of the bias is due to the assumption made in (7.15). The MLE given by (7.17) is actually valid only for large N. To find the exact MLE we would have to minimize J as given by (7.13) by evaluating it for all ϕ. This could be done by using a grid search to find the minimum. Also, observe from the table that the variance for $N = 20$ is below the CRLB. This is possible due to the bias of the estimator, thereby invalidating the CRLB which assumes an unbiased estimator.

Next we fixed the data record length at $N = 80$ and varied the SNR. Plots of the mean and variance versus the SNR, as well as the asymptotic values, are shown in Figures 7.3 and 7.4. As shown in Figure 7.3, the estimator attains the asymptotic mean above about -10 dB. In Figure 7.4 we have plotted $10\log_{10}\mathrm{var}(\hat{\phi})$. This has the desirable effect of causing the CRLB to be a straight line when plotted versus the SNR in dB. In particular, from (7.19) the asymptotic variance or CRLB is

$$\begin{aligned} 10\log_{10}\mathrm{var}(\hat{\phi}) &= 10\log_{10}\frac{1}{N\eta} \\ &= -10\log_{10}N - 10\log_{10}\eta. \end{aligned}$$

Examining the results, we see a peculiar trend. For low SNRs the variance is higher than the CRLB. Only at higher SNRs is the CRLB attained. Hence, the required data record length for the asymptotic results to apply also depends on the SNR. To

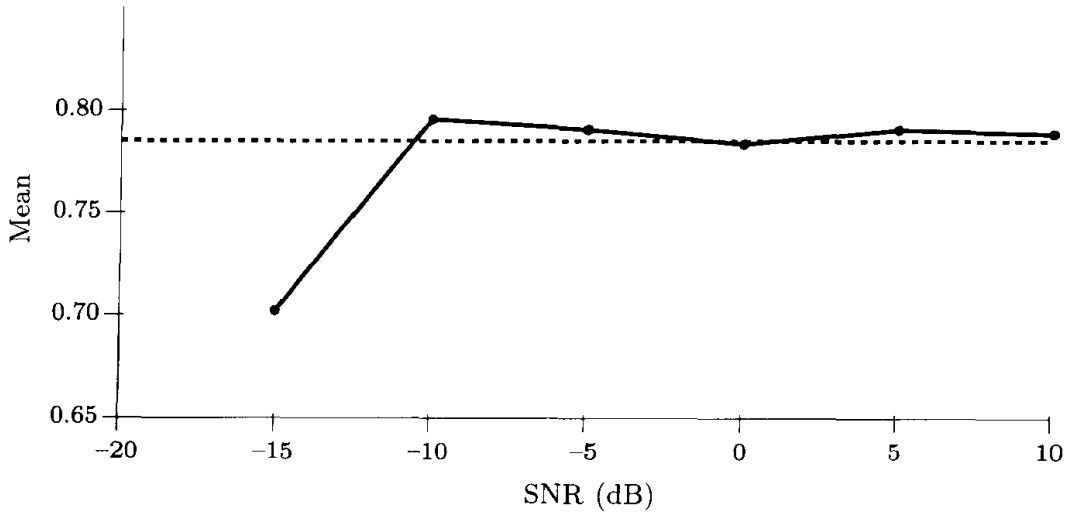


Figure 7.3 Actual vs. asymptotic mean for phase estimator

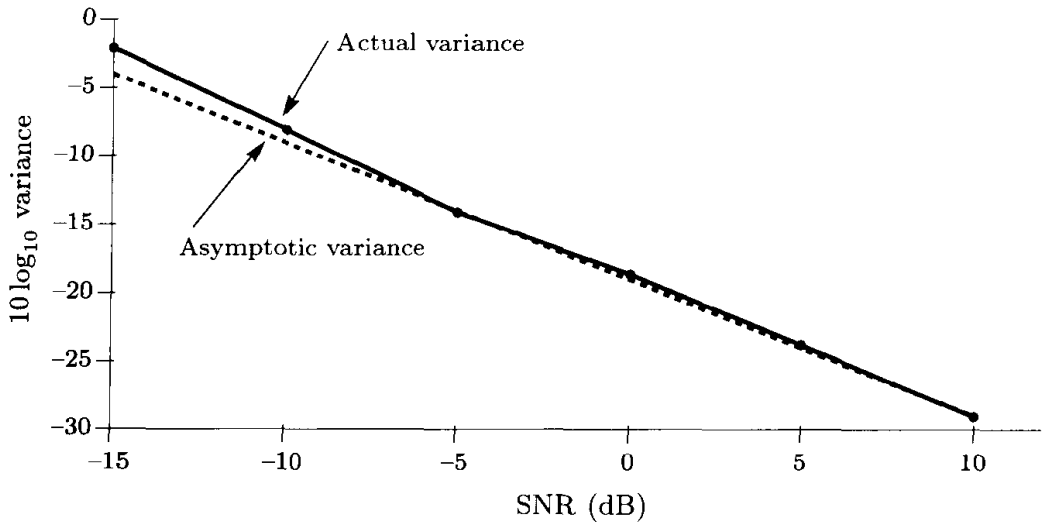
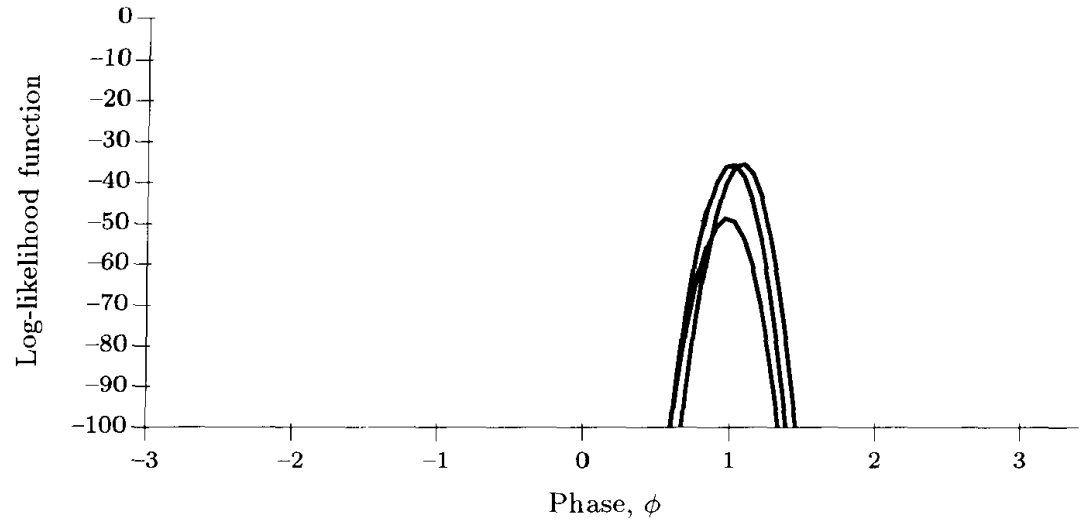
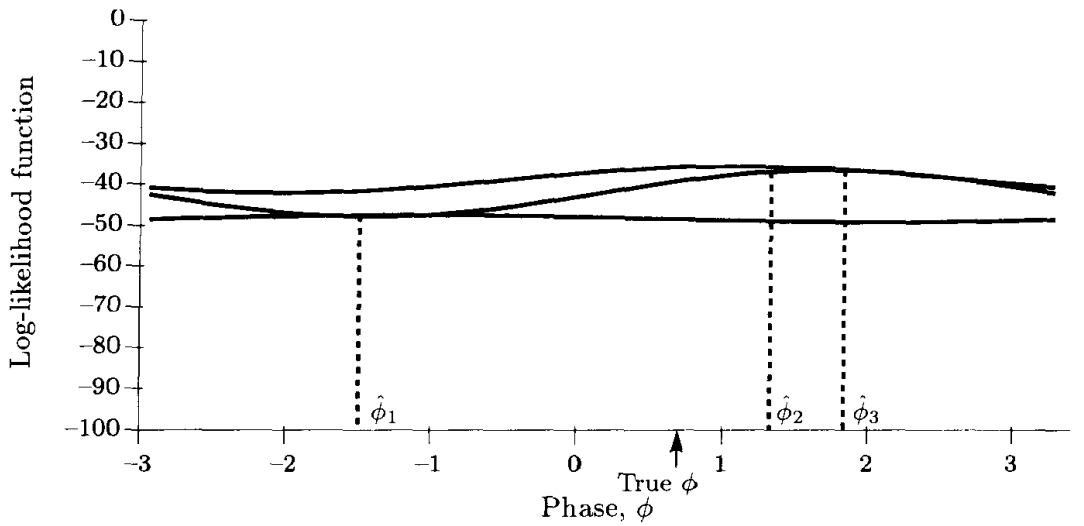


Figure 7.4 Actual vs. asymptotic variance for phase estimator

understand why this occurs we plot typical realizations of the log-likelihood function for different SNRs. As seen in Figure 7.5 for a high SNR, the maximum is relatively stable from realization to realization, and hence the MLE exhibits a low variance. For lower SNRs, however, the effect of the increased noise is to cause other peaks to occur. Occasionally these peaks are larger than the peak near the true value, causing a large estimation error and ultimately a larger variance. These large error estimates are said to be *outliers* and cause the *threshold* effect seen in Figures 7.3 and 7.4. Nonlinear estimators nearly always exhibit this effect. ◇



(a) High SNR (10 dB)



(b) Low SNR (-15 dB)

Figure 7.5 Typical realizations of log-likelihood function for phase

In summary, the asymptotic PDF of the MLE is valid for large enough data records. For signal in noise problems the CRLB may be attained even for short data records if the SNR is high enough. To see why this is so the phase estimator can be written from (7.17) as

$$\hat{\phi} = -\arctan \frac{\sum_{n=0}^{N-1} [A\cos(2\pi f_0 n + \phi) + w[n]] \sin 2\pi f_0 n}{\sum_{n=0}^{N-1} [A\cos(2\pi f_0 n + \phi) + w[n]] \cos 2\pi f_0 n}$$

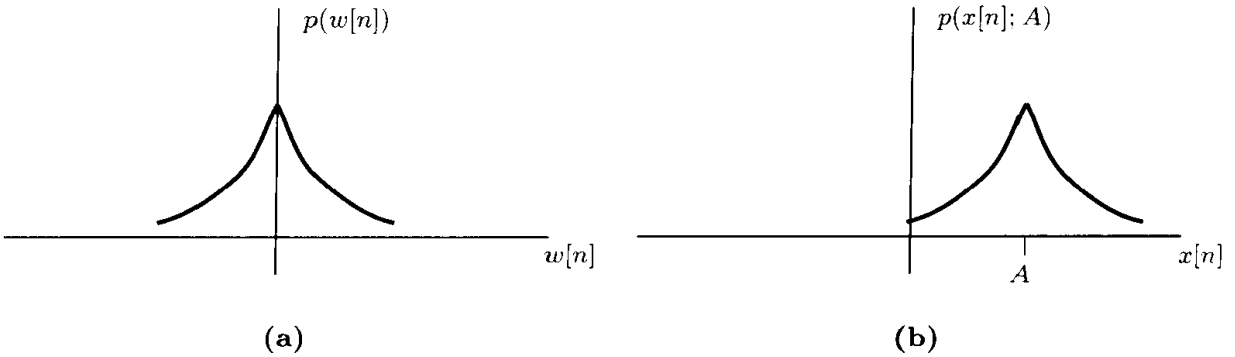


Figure 7.6 Non-Gaussian PDF for Example 7.7

$$\approx -\arctan \frac{-\frac{NA}{2}\sin\phi + \sum_{n=0}^{N-1} w[n]\sin 2\pi f_0 n}{\frac{NA}{2}\cos\phi + \sum_{n=0}^{N-1} w[n]\cos 2\pi f_0 n}$$

where we have used the same type of approximation as in (7.15) and some standard trigonometric identities. Simplifying, we have

$$\hat{\phi} \approx \arctan \frac{\sin\phi - \frac{2}{NA}\sum_{n=0}^{N-1} w[n]\sin 2\pi f_0 n}{\cos\phi + \frac{2}{NA}\sum_{n=0}^{N-1} w[n]\cos 2\pi f_0 n}. \tag{7.20}$$

If the data record is large and/or the sinusoidal power is large, the noise terms will be small. *It is this condition, that the estimation error is small, that allows the MLE to attain its asymptotic distribution.* See also Problem 7.15 for a further discussion of this point.

In some cases the asymptotic distribution does not hold, no matter how large the data record and/or the SNR becomes. This tends to occur when the estimation error cannot be reduced due to a lack of averaging in the estimator. An example follows.

Example 7.7 - DC Level in Nonindependent Non-Gaussian Noise

Consider the observations

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where each sample of $w[n]$ has the PDF $p(w[n])$ as shown in Figure 7.6a. The PDF is symmetric about $w[n] = 0$ and has a maximum at $w[n] = 0$. Furthermore, we assume all the noise samples are equal or $w[0] = w[1] = \cdots = w[N-1]$. In estimating A we need to consider only a single observation since all observations are identical. Utilizing

only $x[0]$, we first note that the PDF of $x[0]$ is a shifted version of $p(w[n])$, where the shift is A as shown in Figure 7.6b. This is because $p_{x[0]}(x[0];A) = p_{w[0]}(x[0]-A)$. The MLE of A is the value that maximizes $p_{w[0]}(x[0]-A)$, which because the PDF of $w[0]$ has a maximum at $w[0]=0$ becomes

$$\hat{A} = x[0].$$

This estimator has the mean

$$E(\hat{A}) = E(x[0]) = A$$

since the noise PDF is symmetric about $w[0]=0$. The variance of $\hat{A}$ is the same as the variance of $x[0]$ or of $w[0]$. Hence,

$$\text{var}(\hat{A}) = \int_{-\infty}^{\infty} u^2 p_{w[0]}(u)\,du$$

while the CRLB, is from Problem 3.2,

$$\text{var}(\hat{A}) \geq \left[\int_{-\infty}^{\infty} \frac{\left(\frac{dp_{w[0]}(u)}{du}\right)^2}{p_{w[0]}(u)} du \right]^{-1}$$

and the two are not in general equal (see Problem 7.16). In this example, then, the estimation error does not decrease as the data record length increases but remains the same. Furthermore, the PDF of $\hat{A} = x[0]$ as shown in Figure 7.6b is a shifted version of $p(w[n])$, clearly not Gaussian. Finally, $\hat{A}$ is not even consistent as $N \to \infty$. $\diamond$

7.6 MLE for Transformed Parameters

In many instances we wish to estimate a *function* of θ, the parameter characterizing the PDF. For example, we may not be interested in the value of a DC level A in WGN but only in the power A^2. In such a situation the MLE of A^2 is easily found from the MLE of A. Some examples illustrate how this is done.

Example 7.8 - Transformed DC Level in WGN

Consider the data

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN with variance σ^2. We wish to find the MLE of $\alpha = \exp(A)$. The PDF is given as

$$p(\mathbf{x};A) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]-A)^2\right] \qquad -\infty < A < \infty \qquad (7.21)$$

and is parameterized by the parameter $\theta = A$. However, since α is a one-to-one transformation of A, we can equivalently parameterize the PDF as

$$p_T(\mathbf{x};\alpha) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]-\ln\alpha)^2\right] \quad \alpha > 0 \tag{7.22}$$

where the subscript T indicates that the PDF is parameterized according to the *transformed* parameter. Clearly, $p_T(\mathbf{x};\alpha)$ is the PDF of the data set

$$x[n] = \ln\alpha + w[n] \qquad n = 0, 1, \ldots, N-1$$

and (7.21) and (7.22) are entirely equivalent. The MLE of α is found by maximizing (7.22) over α. Setting the derivative of $p_T(\mathbf{x};\alpha)$ with respect to α equal to zero yields

$$\sum_{n=0}^{N-1}(x[n]-\ln\hat{\alpha})\frac{1}{\hat{\alpha}} = 0$$

or

$$\hat{\alpha} = \exp(\bar{x}).$$

But $\bar{x}$ is just the MLE of A, so that

$$\hat{\alpha} = \exp(\hat{A}) = \exp(\hat{\theta}).$$

The MLE of the transformed parameter is found by substituting the MLE of the original parameter into the transformation. This property of the MLE is termed the *invariance property.*

Example 7.9 - Transformed DC Level in WGN (Another Example)

Now consider the transformation $\alpha = A^2$ for the data set in the previous example. If we try to repeat the steps, we soon encounter a problem. Attempting to parameterize $p(\mathbf{x};A)$ with respect to α, we find that

$$A = \pm\sqrt{\alpha}$$

since the transformation is not one-to-one. If we choose $A = \sqrt{\alpha}$, then some of the possible PDFs of (7.21) will be missing. We actually require two sets of PDFs

$$\begin{aligned} p_{T_1}(\mathbf{x};\alpha) &= \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]-\sqrt{\alpha})^2\right] \quad \alpha \geq 0 \\ p_{T_2}(\mathbf{x};\alpha) &= \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]+\sqrt{\alpha})^2\right] \quad \alpha > 0 \end{aligned} \tag{7.23}$$

to characterize all possible PDFs. It is possible to find the MLE of α as the value of α that yields the maximum of $p_{T_1}(\mathbf{x};\alpha)$ and $p_{T_2}(\mathbf{x};\alpha)$ or

$$\hat{\alpha} = \arg\max_{\alpha}\left\{p_{T_1}(\mathbf{x};\alpha), p_{T_2}(\mathbf{x};\alpha)\right\}. \tag{7.24}$$

Alternatively, we can find the maximum in two steps as

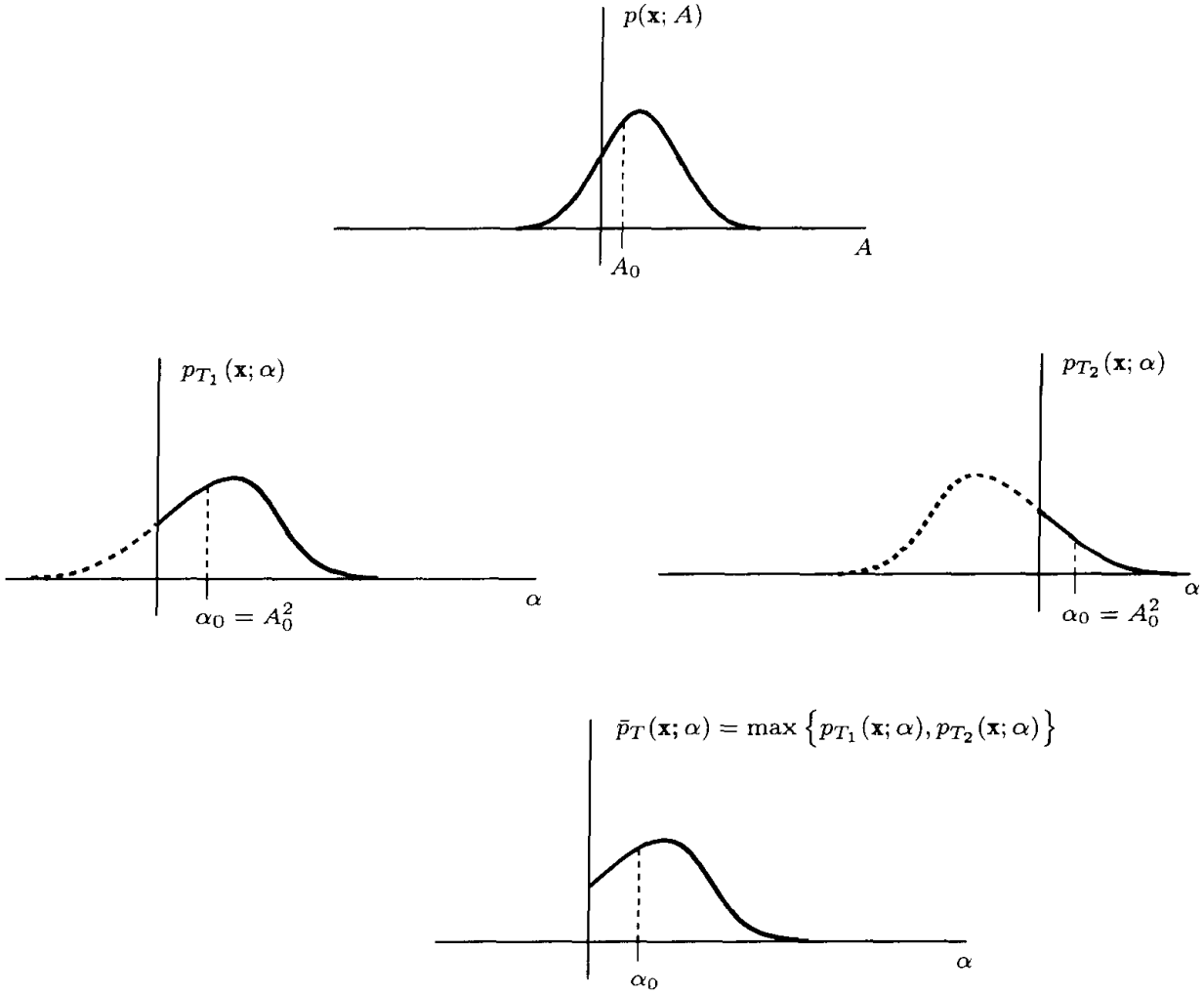


Figure 7.7 Construction of modified likelihood function

1. For a given value of α, say α_0, determine whether $p_{T_1}(\mathbf{x}; \alpha)$ or $p_{T_2}(\mathbf{x}; \alpha)$ is larger. If, for example,

$$p_{T_1}(\mathbf{x}; \alpha_0) > p_{T_2}(\mathbf{x}; \alpha_0),$$

   then denote the value of $p_{T_1}(\mathbf{x}; \alpha_0)$ as $\bar{p}_T(\mathbf{x}; \alpha_0)$. Repeat for all $\alpha > 0$ to form $\bar{p}_T(\mathbf{x}; \alpha)$. (Note that $\bar{p}_T(\mathbf{x}; \alpha = 0) = p(\mathbf{x}; A = 0)$.)

2. The MLE is given as the α that maximizes $\bar{p}_T(\mathbf{x}; \alpha)$ over $\alpha \geq 0$.

This procedure is illustrated in Figure 7.7. The function $\bar{p}_T(\mathbf{x}; \alpha)$ can be thought of as a *modified likelihood function*, having been derived from the original likelihood function by transforming the value of A that yields the maximum value for a given α. In this example, for each α the possible values of A are $\pm\sqrt{\alpha}$. Now, from (7.24) the MLE $\hat{\alpha}$ is

$$\hat{\alpha} = \arg\max_{\alpha \geq 0} \left\{p(\mathbf{x}; \sqrt{\alpha}), p(\mathbf{x}; -\sqrt{\alpha})\right\}$$

$$\begin{aligned} &= \left[\arg \max_{\sqrt{\alpha}\geq 0} \{p(\mathbf{x};\sqrt{\alpha}), p(\mathbf{x};-\sqrt{\alpha})\}\right]^2 \\ &= \left[\arg \max_{-\infty<A<\infty} p(\mathbf{x};A)\right]^2 \\ &= \hat{A}^2 \\ &= \bar{x}^2 \end{aligned}$$

so that again the invariance property holds. The understanding is that $\hat{\alpha}$ maximizes the *modified* likelihood function $\bar{p}_T(\mathbf{x};\alpha)$ since the standard likelihood function with α as a parameter cannot be defined. ◇

We summarize the preceding discussion in the following theorem.

Theorem 7.2 (Invariance Property of the MLE) *The MLE of the parameter $\alpha = g(\theta)$, where the PDF $p(\mathbf{x};\theta)$ is parameterized by θ, is given by*

$$\hat{\alpha} = g(\hat{\theta})$$

where $\hat{\theta}$ is the MLE of θ. The MLE of $\hat{\theta}$ is obtained by maximizing $p(\mathbf{x};\theta)$. If g is not a one-to-one function, then $\hat{\alpha}$ maximizes the modified likelihood function $\bar{p}_T(\mathbf{x};\alpha)$, defined as

$$\bar{p}_T(\mathbf{x};\alpha) = \max_{\{\theta:\alpha=g(\theta)\}} p(\mathbf{x};\theta).$$

We complete our discussion by giving another example.

Example 7.10 - Power of WGN in dB

We observe N samples of WGN with variance σ^2 whose power in dB is to be estimated. To do so we first find the MLE of σ^2. Then, we use the invariance principle to find the power P in dB, which is defined as

$$P = 10\log_{10}\sigma^2.$$

The PDF is given by

$$p(\mathbf{x};\sigma^2) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1} x^2[n]\right].$$

Differentiating the log-likelihood function produces

$$\begin{aligned} \frac{\partial \ln p(\mathbf{x};\sigma^2)}{\partial\sigma^2} &= \frac{\partial}{\partial\sigma^2}\left[-\frac{N}{2}\ln 2\pi - \frac{N}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_{n=0}^{N-1} x^2[n]\right] \\ &= -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{n=0}^{N-1} x^2[n] \end{aligned}$$

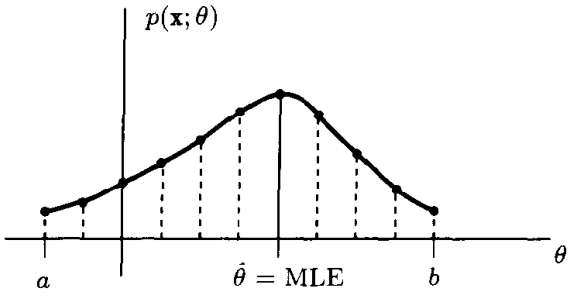


Figure 7.8 Grid search for MLE

and upon setting it equal to zero yields the MLE

$$\hat{\sigma^2} = \frac{1}{N}\sum_{n=0}^{N-1} x^2[n].$$

The MLE of the power in dB readily follows as

$$\begin{aligned}\hat{P} &= 10\log_{10}\hat{\sigma^2} \\ &= 10\log_{10}\frac{1}{N}\sum_{n=0}^{N-1} x^2[n].\end{aligned}$$

◇

7.7 Numerical Determination of the MLE

A distinct advantage of the MLE is that we can always find it for a given data set *numerically.* This is because the MLE is determined as the maximum of a known function, namely, the likelihood function. If, for example, the allowable values of θ lie in the interval $[a, b]$, then we need only maximize $p(\mathbf{x};\theta)$ over that interval. The "safest" way to do this is to perform a grid search over the $[a, b]$ interval as shown in Figure 7.8. As long as the spacing between θ values is small enough, we are guaranteed to find the MLE for the given set of data. Of course, for a new set of data we will have to repeat the search since the likelihood function will undoubtedly change. If, however, the range of θ is not confined to a finite interval, as in estimating the variance of a noise process for which $\sigma^2 > 0$, then a grid search may not be computationally feasible. In such a case, we are forced to resort to iterative maximization procedures. Some typical ones are the Newton-Raphson method, the scoring approach, and the expectation-maximization algorithm. In general, these methods will produce the MLE if the initial guess is close to the true maximum. If not, convergence may not be attained, or only convergence to a local maximum. The difficulty with the use of these iterative methods is that in general we do not know beforehand if they will converge and, even if convergence is attained, whether the value produced is the MLE. An important distinction of our estimation problem which sets it apart from other maximization problems is that the function to be maximized is *not known a priori.* The likelihood function changes for each data set,

requiring the maximization of a *random function.* Nevertheless, these methods can at times produce good results. We now describe some of the more common ones. The interested reader is referred to [Bard 1974] for a more complete description of methods for nonlinear optimization as applied to estimation problems.

As a means of comparison, we will apply the methods to the following example.

Example 7.11 - Exponential in WGN

Consider the data

$$x[n] = r^n + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN with variance σ^2. The parameter r, the exponential factor, is to be estimated. Allowable values are $r > 0$. The MLE of r is the value that maximizes the likelihood function

$$p(\mathbf{x}; r) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n] - r^n)^2\right]$$

or, equivalently, the value that minimizes

$$J(r) = \sum_{n=0}^{N-1}(x[n] - r^n)^2.$$

Differentiating $J(r)$ and setting it equal to zero produces

$$\sum_{n=0}^{N-1}(x[n] - r^n)\, nr^{n-1} = 0. \tag{7.25}$$

This is a nonlinear equation in r and cannot be solved directly. We will now consider the iterative methods of Newton-Raphson and scoring. ◇

The iterative methods attempt to maximize the log-likelihood function by finding a zero of the derivative function. To do so the derivative is taken and set equal to zero, yielding

$$\frac{\partial \ln p(\mathbf{x}; \theta)}{\partial \theta} = 0. \tag{7.26}$$

Then, the methods attempt to solve this equation iteratively. Let

$$g(\theta) = \frac{\partial \ln p(\mathbf{x}; \theta)}{\partial \theta}$$

and assume that we have an initial guess for the solution to (7.26). Call this guess θ_0. Then, if $g(\theta)$ is approximately linear near θ_0, we can approximate it by

$$g(\theta) \approx g(\theta_0) + \left.\frac{dg(\theta)}{d\theta}\right|_{\theta=\theta_0}(\theta - \theta_0) \tag{7.27}$$

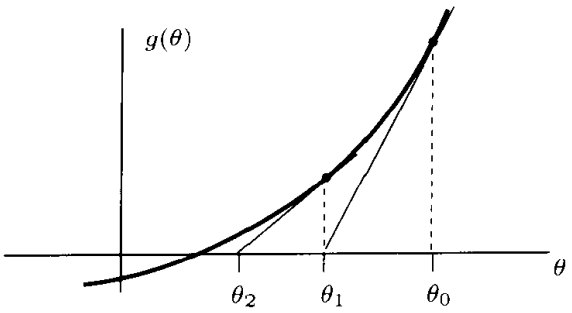


Figure 7.9 Newton-Raphson method for finding zero of function

as shown in Figure 7.9. Next, we use (7.27) to solve for the zero θ_1, so that upon setting $g(\theta_1)$ equal to zero and solving for θ_1 we have

$$\theta_1 = \theta_0 - \frac{g(\theta_0)}{\left.\frac{dg(\theta)}{d\theta}\right|_{\theta=\theta_0}}.$$

Again we linearize g but use the new guess, θ_1, as our point of linearization and repeat the previous procedure to find the new zero. As shown in Figure 7.9, the sequence of guesses will converge to the true zero of $g(\theta)$. In general, the Newton-Raphson iteration finds the new guess, θ_{k+1}, based on the previous one, θ_k, using

$$\theta_{k+1} = \theta_k - \frac{g(\theta_k)}{\left.\frac{dg(\theta)}{d\theta}\right|_{\theta=\theta_k}}. \tag{7.28}$$

Note that at convergence $\theta_{k+1} = \theta_k$, and from (7.28) $g(\theta_k) = 0$, as desired. Since $g(\theta)$ is the derivative of the log-likelihood function, we find the MLE as

$$\theta_{k+1} = \theta_k - \left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2}\right]^{-1} \left.\frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta}\right|_{\theta=\theta_k}. \tag{7.29}$$

Several points need to be raised concerning the Newton-Raphson iterative procedure.

1. The iteration may not converge. This will be particularly evident when the second derivative of the log-likelihood function is small. In this case it is seen from (7.29) that the correction term may fluctuate wildly from iteration to iteration.

2. Even if the iteration converges, the point found may not be the global maximum but possibly only a local maximum or even a local minimum. Hence, to avoid these possibilities it is best to use several starting points and at convergence choose the one that yields the maximum. Generally, if the initial point is close to the global maximum, the iteration will converge to it. *The importance of a good initial guess cannot be overemphasized.* An illustration is given in Problem 7.18.

Applying the Newton-Raphson iteration to the problem in Example 7.11, we have

$$\begin{aligned}\frac{\partial \ln p(\mathbf{x};r)}{\partial r} &= \frac{1}{\sigma^2}\sum_{n=0}^{N-1}(x[n]-r^n)nr^{n-1} \\ \frac{\partial^2 \ln p(\mathbf{x};r)}{\partial r^2} &= \frac{1}{\sigma^2}\left[\sum_{n=0}^{N-1}n(n-1)x[n]r^{n-2}-\sum_{n=0}^{N-1}n(2n-1)r^{2n-2}\right] \\ &= \frac{1}{\sigma^2}\sum_{n=0}^{N-1}nr^{n-2}\left[(n-1)x[n]-(2n-1)r^n\right] \end{aligned} \tag{7.30}$$

and thus the Newton-Raphson iteration becomes

$$r_{k+1} = r_k - \frac{\displaystyle\sum_{n=0}^{N-1}(x[n]-r_k^n)nr_k^{n-1}}{\displaystyle\sum_{n=0}^{N-1}nr_k^{n-2}\left[(n-1)x[n]-(2n-1)r_k^n\right]}. \tag{7.31}$$

A second common iterative procedure is the method of scoring. It recognizes that

$$\left.\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2}\right|_{\theta=\theta_k} \approx -I(\theta_k) \tag{7.32}$$

where $I(\theta)$ is the Fisher information. Indeed, for IID samples we have

$$\begin{aligned}\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2} &= \sum_{n=0}^{N-1}\frac{\partial^2 \ln p(x[n];\theta)}{\partial \theta^2} \\ &= N\frac{1}{N}\sum_{n=0}^{N-1}\frac{\partial^2 \ln p(x[n];\theta)}{\partial \theta^2} \\ &\approx NE\left[\frac{\partial^2 \ln p(x[n];\theta)}{\partial \theta^2}\right] \\ &= -Ni(\theta) \\ &= -I(\theta)\end{aligned}$$

by the law of large numbers. Presumably, replacing the second derivative by its expected value will increase the stability of the iteration. The method then becomes

$$\theta_{k+1} = \theta_k + I^{-1}(\theta)\left.\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right|_{\theta=\theta_k}. \tag{7.33}$$

This approach is termed the *method of scoring*. It too suffers from the same convergence problems as the Newton-Raphson iteration. As applied to the problem in Example 7.11,

TABLE 7.3 Sequence of Iterates for Newton-Raphson Method

Iteration	Initial Guess, r_0		
	0.8	0.2	1.2
1	0.723	0.799	1.187
2	0.638	0.722	1.174
3	0.561	0.637	1.161
4	0.510	0.560	1.148
5	0.494	0.510	1.136
6	0.493	0.494	1.123
7		0.493	1.111
8			1.098
9			1.086
10			1.074
⋮			⋮
29			0.493

it produces upon using (7.30)

$$I(\theta) = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} n^2 r^{2n-2}$$

so that

$$r_{k+1} = r_k + \frac{\sum_{n=0}^{N-1} (x[n] - r_k^n) n r_k^{n-1}}{\sum_{n=0}^{N-1} n^2 r_k^{2n-2}}. \tag{7.34}$$

As an example of the iterative approach, we implement the Newton-Raphson method for Example 7.11 using a computer simulation. Using $N = 50$, $r = 0.5$, and $\sigma^2 = 0.01$, we generated a realization of the process. For a particular outcome $-J(r)$ is plotted in Figure 7.10 for $0 < r < 1$. The peak of the function, which is the MLE, occurs at $r = 0.493$. It is seen that the function is fairly broad for $r < 0.5$ but rather sharp for larger values of r. In fact, for $r > 1$ it was difficult to plot due to the exponential signal r^n, which causes the function to become very large and negative. We applied the Newton-Raphson method as given by (7.31) using several initial guesses. The resulting iterates are listed in Table 7.3. For $r_0 = 0.8$ or $r_0 = 0.2$ the iteration quickly converged to the true maximum. However, for $r_0 = 1.2$ the convergence was much slower, although the true maximum was attained after 29 iterations. If the initial guess was less than about 0.18 or greater than about 1.2, the succeeding iterates exceeded 1 and kept

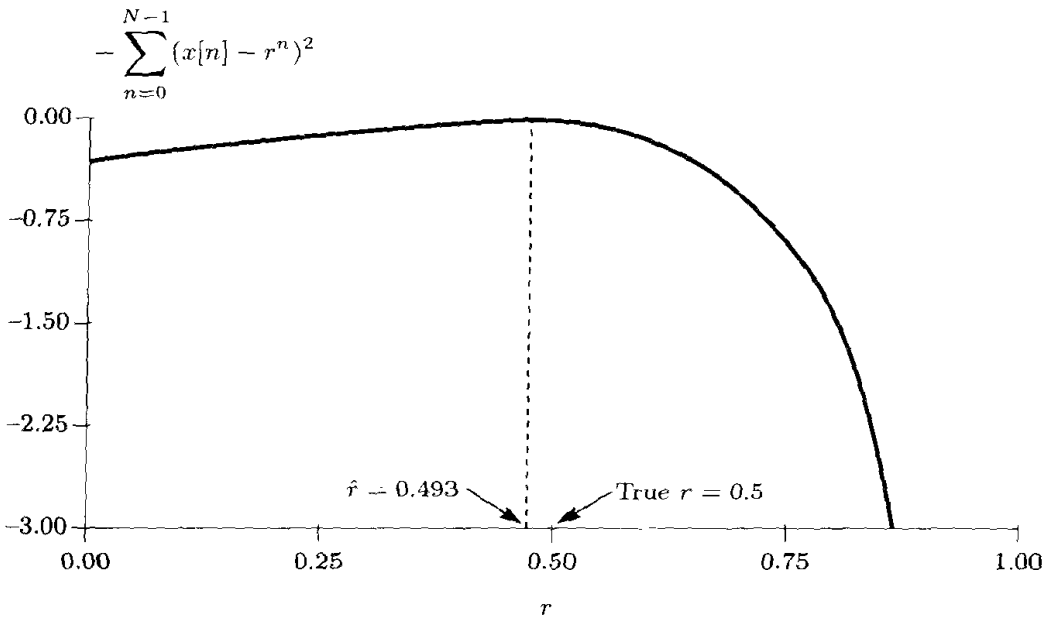


Figure 7.10 Function to be maximized for MLE

increasing, causing a computer overflow condition in the algorithm. In this case the Newton-Raphson method fails to converge. As expected, the method works well when the initial guess is close to the true MLE but may produce poor results otherwise.

A third method that has recently been proposed is the expectation-maximization algorithm. Unlike the first two methods, it requires some ingenuity on the part of the designer. Its advantage, however, is its guaranteed convergence to at least a local maximum under some mild conditions. Convergence to the global maximum is, as with all iterative schemes, not assured. The expectation-maximization algorithm, not being a "turn-the-crank" procedure, does not lend itself to all estimation problems. A particular class of problems for which it is useful involves a vector parameter, and so we defer our discussion until the next section.

7.8 Extension to a Vector Parameter

The MLE for a vector parameter $\boldsymbol{\theta}$ is defined to be the value that maximizes the likelihood function $p(\mathbf{x};\boldsymbol{\theta})$ over the allowable domain for $\boldsymbol{\theta}$. Assuming a differentiable likelihood function, the MLE is found from

$$\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \mathbf{0}. \tag{7.35}$$

If multiple solutions exist, then the one that maximizes the likelihood function is the MLE. We now find the MLE for the problem in Example 3.6.

Example 7.12 - DC Level in WGN

Consider the data

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN with variance σ^2 and the vector parameter $\boldsymbol{\theta} = [A\ \sigma^2]^T$ is to be estimated. Then, from Example 3.6

$$\begin{aligned} \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial A} &= \frac{1}{\sigma^2}\sum_{n=0}^{N-1}(x[n]-A) \\ \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \sigma^2} &= -\frac{N}{2\sigma^2}+\frac{1}{2\sigma^4}\sum_{n=0}^{N-1}(x[n]-A)^2. \end{aligned}$$

Solving for A from the first equation, we have the MLE

$$\hat{A} = \bar{x}.$$

Solving for σ^2 in the second equation and using $\hat{A} = \bar{x}$ produces

$$\hat{\sigma^2} = \frac{1}{N}\sum_{n=0}^{N-1}(x[n]-\bar{x})^2.$$

The MLE is

$$\hat{\boldsymbol{\theta}} = \begin{bmatrix} \bar{x} \\ \frac{1}{N}\sum_{n=0}^{N-1}(x[n]-\bar{x})^2 \end{bmatrix}.$$

◇

For a vector parameter the asymptotic properties of the MLE are summarized in the following theorem.

Theorem 7.3 (Asymptotic Properties of the MLE (Vector Parameter)) *If the PDF $p(\mathbf{x};\boldsymbol{\theta})$ of the data $\mathbf{x}$ satisfies some "regularity" conditions, then the MLE of the unknown parameter $\boldsymbol{\theta}$ is asymptotically distributed according to*

$$\hat{\boldsymbol{\theta}} \overset{a}{\sim} \mathcal{N}(\boldsymbol{\theta}, \mathbf{I}^{-1}(\boldsymbol{\theta})) \tag{7.36}$$

where $\mathbf{I}(\boldsymbol{\theta})$ is the Fisher information matrix evaluated at the true value of the unknown parameter.

The regularity conditions are similar to those discussed in Appendix 7B for the scalar parameter case. As an illustration, consider the previous example. It can be shown that [Hoel, Port, and Stone 1971]

$$\begin{aligned} \bar{x} &\sim \mathcal{N}(A, \sigma^2/N) \\ T = \sum_{n=0}^{N-1}\frac{(x[n]-\bar{x})^2}{\sigma^2} &\sim \chi^2_{N-1} \end{aligned} \tag{7.37}$$

and that the two statistics are independent. For large N it is well known from the central limit theorem that

$$\chi^2_N \overset{a}{\sim} \mathcal{N}(N, 2N)$$

so that

$$T \overset{a}{\sim} \mathcal{N}(N-1, 2(N-1))$$

or

$$\hat{\sigma^2} = \frac{1}{N}\sum_{n=0}^{N-1}(x[n]-\bar{x})^2 \overset{a}{\sim} \mathcal{N}\left(\frac{(N-1)\sigma^2}{N}, \frac{2(N-1)\sigma^4}{N^2}\right). \tag{7.38}$$

The asymptotic PDF of $\hat{\boldsymbol{\theta}}$ is jointly Gaussian since the statistics are individually Gaussian and independent. Furthermore, from (7.37) and (7.38) for large N

$$E(\hat{\boldsymbol{\theta}}) = \begin{bmatrix} A \\ \frac{(N-1)\sigma^2}{N} \end{bmatrix} \rightarrow \begin{bmatrix} A \\ \sigma^2 \end{bmatrix} = \boldsymbol{\theta}$$

$$\mathbf{C}(\boldsymbol{\theta}) = \begin{bmatrix} \frac{\sigma^2}{N} & 0 \\ 0 & \frac{2(N-1)\sigma^4}{N^2} \end{bmatrix} \rightarrow \begin{bmatrix} \frac{\sigma^2}{N} & 0 \\ 0 & \frac{2\sigma^4}{N} \end{bmatrix} = \mathbf{I}^{-1}(\boldsymbol{\theta})$$

where the inverse of the Fisher information matrix has previously been given in Example 3.6. Hence, the asymptotic distribution is described by (7.36).

In some instances, the asymptotic PDF of the MLE is not given by (7.36). These situations generally occur when the number of parameters to be estimated is too large relative to the number of data samples available. As in Example 7.7, this situation restricts the averaging in the estimator.

Example 7.13 - Signal in Non-Gaussian Noise

Consider the data

$$x[n] = s[n] + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is zero mean IID noise with the Laplacian PDF

$$p(w[n]) = \frac{1}{4}\exp\left[-\frac{1}{2}|w[n]|\right].$$

The signal samples $\{s[0], s[1], \ldots, s[N-1]\}$ are to be estimated. The PDF of the data is

$$p(\mathbf{x}; \boldsymbol{\theta}) = \prod_{n=0}^{N-1}\frac{1}{4}\exp\left[-\frac{1}{2}|x[n]-s[n]|\right]. \tag{7.39}$$

The MLE of $\boldsymbol{\theta} = [s[0]\, s[1] \ldots s[N-1]]^T$ is easily seen to be

$$\hat{s}[n] = x[n] \qquad n = 0, 1, \ldots, N-1$$

or, equivalently,

$$\hat{\boldsymbol{\theta}} = \mathbf{x}.$$

It is clear that the MLE is *not* Gaussian even as $N \to \infty$. The PDF of $\hat{\boldsymbol{\theta}}$ is given by (7.39) with $\mathbf{x}$ replaced by $\hat{\boldsymbol{\theta}}$. The difficulty is that no averaging is possible since we have chosen to estimate as many parameters as data points. Therefore, the central limit theorem, which accounts for the asymptotic Gaussian PDF of the MLE (see Appendix 7B), does not apply. ◇

As in the scalar case, the invariance property holds, as summarized by the following theorem.

Theorem 7.4 (Invariance Property of MLE (Vector Parameter)) *The MLE of the parameter $\boldsymbol{\alpha} = \mathbf{g}(\boldsymbol{\theta})$, where $\mathbf{g}$ is an r-dimensional function of the $p \times 1$ parameter $\boldsymbol{\theta}$, and the PDF $p(\mathbf{x};\boldsymbol{\theta})$ is parameterized by $\boldsymbol{\theta}$, is given by*

$$\hat{\boldsymbol{\alpha}} = \mathbf{g}(\hat{\boldsymbol{\theta}})$$

for $\hat{\boldsymbol{\theta}}$, the MLE of $\boldsymbol{\theta}$. If $\mathbf{g}$ is not an invertible function, then $\hat{\boldsymbol{\alpha}}$ maximizes the modified likelihood function $\bar{p}_T(\mathbf{x};\boldsymbol{\alpha})$, defined as

$$\bar{p}_T(\mathbf{x};\boldsymbol{\alpha}) = \max_{\{\boldsymbol{\theta}:\boldsymbol{\alpha}=\mathbf{g}(\boldsymbol{\theta})\}} p(\mathbf{x};\boldsymbol{\theta}).$$

An example of the use of this theorem can be found in Problem 7.21.

In Chapter 3 we discussed the computation of the CRLB for the general Gaussian case in which the observed data vector had the PDF

$$\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}(\boldsymbol{\theta}), \mathbf{C}(\boldsymbol{\theta})).$$

In Appendix 3C (see (3C.5)) it was shown that the partial derivatives of the log-likelihood function were

$$\begin{aligned}\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_k} &= -\frac{1}{2}\mathrm{tr}\left(\mathbf{C}^{-1}(\boldsymbol{\theta})\frac{\partial \mathbf{C}(\boldsymbol{\theta})}{\partial \theta_k}\right) + \frac{\partial \boldsymbol{\mu}(\boldsymbol{\theta})^T}{\partial \theta_k}\mathbf{C}^{-1}(\boldsymbol{\theta})(\mathbf{x}-\boldsymbol{\mu}(\boldsymbol{\theta})) \\ &\quad -\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}(\boldsymbol{\theta}))^T\frac{\partial \mathbf{C}^{-1}(\boldsymbol{\theta})}{\partial \theta_k}(\mathbf{x}-\boldsymbol{\mu}(\boldsymbol{\theta})) \end{aligned} \tag{7.40}$$

for $k = 1, 2, \ldots, p$. By setting (7.40) equal to zero we can obtain necessary conditions for the MLE and on occasion, if solvable, the exact MLE. A somewhat more convenient form uses (3C.2) in Appendix 3C to replace $\partial \mathbf{C}^{-1}(\boldsymbol{\theta})/\partial \theta_k$ in (7.40). An important example of the latter is the linear model described in detail in Chapter 4. Recall that the general linear data model is

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w} \tag{7.41}$$

where $\mathbf{H}$ is a known $N \times p$ matrix and $\mathbf{w}$ is a noise vector of dimension $N \times 1$ with PDF $\mathcal{N}(\mathbf{0}, \mathbf{C})$. Under these conditions the PDF is

$$p(\mathbf{x};\boldsymbol{\theta}) = \frac{1}{(2\pi)^{\frac{N}{2}}\det^{\frac{1}{2}}(\mathbf{C})}\exp\left[-\frac{1}{2}(\mathbf{x}-\mathbf{H}\boldsymbol{\theta})^T\mathbf{C}^{-1}(\mathbf{x}-\mathbf{H}\boldsymbol{\theta})\right]$$

so that the MLE of $\boldsymbol{\theta}$ is found by minimizing

$$J(\boldsymbol{\theta}) = (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T \mathbf{C}^{-1}(\mathbf{x} - \mathbf{H}\boldsymbol{\theta}). \tag{7.42}$$

Since this is a quadratic function of the elements of $\boldsymbol{\theta}$ and $\mathbf{C}^{-1}$ is a positive definite matrix, differentiation will produce the global minimum. Now, using (7.40) and noting that

$$\boldsymbol{\mu}(\boldsymbol{\theta}) = \mathbf{H}\boldsymbol{\theta}$$

and that the covariance matrix does not depend on $\boldsymbol{\theta}$, we have

$$\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_k} = \frac{\partial (\mathbf{H}\boldsymbol{\theta})^T}{\partial \theta_k} \mathbf{C}^{-1}(\mathbf{x} - \mathbf{H}\boldsymbol{\theta}).$$

Combining the partial derivatives to form the gradient, we have

$$\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \frac{\partial (\mathbf{H}\boldsymbol{\theta})^T}{\partial \boldsymbol{\theta}} \mathbf{C}^{-1}(\mathbf{x} - \mathbf{H}\boldsymbol{\theta})$$

so that upon setting the gradient equal to zero we have

$$\mathbf{H}^T \mathbf{C}^{-1}(\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}) = \mathbf{0}.$$

Solving for $\hat{\boldsymbol{\theta}}$ produces the MLE

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{C}^{-1} \mathbf{x}. \tag{7.43}$$

But this estimator was shown in Chapter 4 to be the MVU estimator as well as an efficient estimator. As such, $\hat{\boldsymbol{\theta}}$ is unbiased and has a covariance of

$$\mathbf{C}_{\hat{\theta}} = (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1}. \tag{7.44}$$

Finally, the PDF of the MLE is Gaussian (being a linear function of $\mathbf{x}$), so that the asymptotic properties of Theorem 7.3 are attained even for finite data records. For the linear model it can be said that *the MLE is optimal.* These results are summarized by the following theorem.

Theorem 7.5 (Optimality of the MLE for the Linear Model) *If the observed data* $\mathbf{x}$ *are described by the general linear model*

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w} \tag{7.45}$$

where $\mathbf{H}$ *is a known* $N \times p$ *matrix with* $N > p$ *and of rank* p, $\boldsymbol{\theta}$ *is a* $p \times 1$ *parameter vector to be estimated, and* $\mathbf{w}$ *is a noise vector with PDF* $\mathcal{N}(\mathbf{0}, \mathbf{C})$, *then the MLE of* $\boldsymbol{\theta}$ *is*

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{C}^{-1} \mathbf{x}. \tag{7.46}$$

$\hat{\boldsymbol{\theta}}$ *is also an efficient estimator in that it attains the CRLB and hence is the MVU estimator. The PDF of* $\hat{\boldsymbol{\theta}}$ *is*

$$\hat{\boldsymbol{\theta}} \sim \mathcal{N}(\boldsymbol{\theta}, (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1}). \tag{7.47}$$

Many examples are given in Chapter 4. The preceding result can be generalized to assert that *if an efficient estimator exists, it is given by the MLE* (see Problem 7.12).

In finding the MLE it is quite common to have to resort to numerical techniques of maximization. The Newton-Raphson and scoring methods were described in Section 7.7. They are easily extended to the vector parameter case. The Newton-Raphson iteration becomes

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \left[\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}^T}\right]^{-1} \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial\boldsymbol{\theta}}\bigg|_{\boldsymbol{\theta}=\boldsymbol{\theta}_k} \tag{7.48}$$

where

$$\left[\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}^T}\right]_{ij} = \frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial\theta_i\partial\theta_j} \qquad \begin{matrix} i = 1,2,\ldots,p \\ j = 1,2,\ldots,p \end{matrix}$$

is the Hessian of the log-likelihood function and $\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial\boldsymbol{\theta}}$ is the $p \times 1$ gradient vector. In implementing (7.48) inversion of the Hessian is not required. Rewriting (7.48) as

$$\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}^T}\bigg|_{\boldsymbol{\theta}=\boldsymbol{\theta}_k} \boldsymbol{\theta}_{k+1} = \frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}^T}\bigg|_{\boldsymbol{\theta}=\boldsymbol{\theta}_k} \boldsymbol{\theta}_k - \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial\boldsymbol{\theta}}\bigg|_{\boldsymbol{\theta}=\boldsymbol{\theta}_k} \tag{7.49}$$

we see that the new iterate, $\boldsymbol{\theta}_{k+1}$, can be found from the previous iterate, $\boldsymbol{\theta}_k$, by solving a set of p simultaneous linear equations.

The scoring method is obtained from the Newton-Raphson method by replacing the Hessian by the negative of the Fisher information matrix to yield

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k + \mathbf{I}^{-1}(\boldsymbol{\theta}) \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial\boldsymbol{\theta}}\bigg|_{\boldsymbol{\theta}=\boldsymbol{\theta}_k} \tag{7.50}$$

As explained previously, the inversion may be avoided by writing (7.50) in the form of (7.49). As in the scalar case, the Newton-Raphson and scoring methods may suffer from convergence problems (see Section 7.7). Care must be exercised in using them. Typically, as the data record becomes large, the log-likelihood function becomes more nearly quadratic near the maximum, and the iterative procedure will produce the MLE.

A third method of numerically determining the MLE is the expectation-maximization (EM) algorithm [Dempster, Laird, and Rubin 1977]. This method, although iterative in nature, is guaranteed under certain mild conditions to converge, and at convergence to produce at least a local maximum. It has the desirable property of increasing the likelihood at each step. The EM algorithm exploits the observation that some data sets may allow easier determination of the MLE than the given one. As an example, consider

$$x[n] = \sum_{i=1}^{p} \cos 2\pi f_i n + w[n] \qquad n = 0,1,\ldots,N-1$$

where $w[n]$ is WGN with variance σ^2 and the frequencies $\mathbf{f} = [f_1\ f_2 \ldots f_p]^T$ are to be estimated. The MLE would require a multidimensional minimization of

$$J(\mathbf{f}) = \sum_{n=0}^{N-1} \left(x[n] - \sum_{i=1}^{p} \cos 2\pi f_i n\right)^2.$$

On the other hand, if the original data could be replaced by the independent data sets

$$y_i[n] = \cos 2\pi f_i n + w_i[n] \qquad \begin{array}{l} i = 1, 2, \ldots, p \\ n = 0, 1, \ldots, N-1 \end{array} \tag{7.51}$$

where $w_i[n]$ is WGN with variance σ_i^2, then the problem would be decoupled. It is easily shown that due to the independence assumption the PDF would factor into the PDF for each data set. Consequently, the MLE of f_i could be obtained from a minimization of

$$J(f_i) = \sum_{n=0}^{N-1} (y_i[n] - \cos 2\pi f_i n)^2 \qquad i = 1, 2, \ldots, p.$$

The original p-dimensional minimization has been reduced to p separate one-dimensional minimizations, in general, an easier problem. The new data set $\{y_1[n], y_2[n], \ldots, y_p[n]\}$ is termed the *complete data* and can be related to the original data as

$$x[n] = \sum_{i=1}^{p} y_i[n] \tag{7.52}$$

if the noise is decomposed as

$$w[n] = \sum_{i=1}^{p} w_i[n].$$

For this decomposition to hold we will assume that the $w_i[n]$ noise processes are independent of each other and

$$\sigma^2 = \sum_{i=1}^{p} \sigma_i^2. \tag{7.53}$$

The key question remains as to how to obtain the complete data from the original or *incomplete* data. The reader should also note that the decomposition is not unique. We could have just as easily hypothesized the complete data to be

$$\begin{aligned} y_1[n] &= \sum_{i=1}^{p} \cos 2\pi f_i n \\ y_2[n] &= w[n] \end{aligned}$$

and thus

$$x[n] = y_1[n] + y_2[n]$$

as a signal and noise decomposition.

In general, we suppose that there is a complete to incomplete data transformation given as

$$\mathbf{x} = \mathbf{g}(\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_M) = \mathbf{g}(\mathbf{y}) \tag{7.54}$$

where in the previous example $M = p$ and the elements of $\mathbf{y}_i$ are given by (7.51). The function $\mathbf{g}$ is a many-to-one transformation. We wish to find the MLE of $\boldsymbol{\theta}$ by

maximizing $\ln p_x(\mathbf{x};\boldsymbol{\theta})$. Finding this too difficult, we instead maximize $\ln p_y(\mathbf{y};\boldsymbol{\theta})$. Since $\mathbf{y}$ is unavailable, we replace the log-likelihood function by its conditional expectation or

$$E_{y|x}[\ln p_y(\mathbf{y};\boldsymbol{\theta})] = \int \ln p_y(\mathbf{y};\boldsymbol{\theta}) p(\mathbf{y}|\mathbf{x};\boldsymbol{\theta})\, d\mathbf{y}. \tag{7.55}$$

Finally, since we need to know $\boldsymbol{\theta}$ to determine $p(\mathbf{y}|\mathbf{x};\boldsymbol{\theta})$ and hence the expected log-likelihood function, we use the current guess. Letting $\boldsymbol{\theta}_k$ denote the kth guess of the MLE of $\boldsymbol{\theta}$, we then have the following iterative algorithm:

Expectation (E): Determine the average log-likelihood of the complete data

$$U(\boldsymbol{\theta}, \boldsymbol{\theta}_k) = \int \ln p_y(\mathbf{y};\boldsymbol{\theta}) p(\mathbf{y}|\mathbf{x};\boldsymbol{\theta}_k)\, d\mathbf{y}. \tag{7.56}$$

Maximization (M): Maximize the average log-likelihood function of the complete data

$$\boldsymbol{\theta}_{k+1} = \arg\max_{\boldsymbol{\theta}} U(\boldsymbol{\theta}, \boldsymbol{\theta}_k). \tag{7.57}$$

At convergence we hopefully will have the MLE. This approach is termed the *EM algorithm*. Applying it to our frequency estimation problem, we have the following iteration as derived in Appendix 7C.

E Step: For $i = 1, 2, \ldots, p$

$$\hat{y}_i[n] = \cos 2\pi f_{i_k} n + \beta_i \left(x[n] - \sum_{i=1}^{p} \cos 2\pi f_{i_k} n\right). \tag{7.58}$$

M Step: For $i = 1, 2, \ldots, p$

$$f_{i_{k+1}} = \arg\max_{f_i} \sum_{n=0}^{N-1} \hat{y}_i[n] \cos 2\pi f_i n \tag{7.59}$$

where the β_i's can be arbitrarily chosen as long as $\sum_{i=1}^{p} \beta_i = 1$.

Note that $\beta_i(x[n] - \sum_{i=1}^{p} \cos 2\pi f_{i_k} n)$ in (7.58) is an estimate of $w_i[n]$. Hence, the algorithm iteratively decouples the original data set into p separate data sets, with each one consisting of a *single* sinusoid in WGN. The maximization given above corresponds to the MLE of a single sinusoid with the data set given by the *estimated* complete data (see Problem 7.19). Good results have been obtained with this approach. Its disadvantages are the difficulty of determining the conditional expectation in closed form and the arbitrariness in the choice of the complete data. Nonetheless, for the Gaussian problem this method can easily be applied. The reader should consult [Feder and Weinstein 1988] for further details.

7.9 Asymptotic MLE

In many cases it is difficult to evaluate the MLE of a parameter whose PDF is Gaussian due to the need to invert a large dimension covariance matrix. For example, if $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{C}(\boldsymbol{\theta}))$, the MLE of $\boldsymbol{\theta}$ is obtained by maximizing

$$p(\mathbf{x};\boldsymbol{\theta}) = \frac{1}{(2\pi)^{\frac{N}{2}} \det^{\frac{1}{2}}(\mathbf{C}(\boldsymbol{\theta}))} \exp\left[-\frac{1}{2}\mathbf{x}^T\mathbf{C}^{-1}(\boldsymbol{\theta})\mathbf{x}\right].$$

If the covariance matrix cannot be inverted in closed form, then a search technique will require inversion of the $N \times N$ matrix for each value of $\boldsymbol{\theta}$ to be searched. An alternative *approximate* method can be applied when $\mathbf{x}$ is data from a zero mean WSS random process, so that the covariance matrix is Toeplitz (see Appendix 1). In such a case, it is shown in Appendix 3D that the asymptotic (for large data records) log-likelihood function is given by (see (3D.6))

$$\ln p(\mathbf{x};\boldsymbol{\theta}) = -\frac{N}{2}\ln 2\pi - \frac{N}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}} \left[\ln P_{xx}(f) + \frac{I(f)}{P_{xx}(f)}\right] df \tag{7.60}$$

where

$$I(f) = \frac{1}{N}\left|\sum_{n=0}^{N-1} x[n]\exp(-j2\pi fn)\right|^2$$

is the periodogram of the data and $P_{xx}(f)$ is the PSD. The dependence of the log-likelihood function on $\boldsymbol{\theta}$ is through the PSD. Differentiation of (7.60) produces the necessary conditions for the MLE

$$\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial\theta_i} = -\frac{N}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}} \left[\frac{1}{P_{xx}(f)} - \frac{I(f)}{P_{xx}^2(f)}\right]\frac{\partial P_{xx}(f)}{\partial\theta_i} df$$

or

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \left[\frac{1}{P_{xx}(f)} - \frac{I(f)}{P_{xx}^2(f)}\right]\frac{\partial P_{xx}(f)}{\partial\theta_i} df = 0. \tag{7.61}$$

The second derivative is given by (3D.7) in Appendix 3D, so that the Newton- Raphson or scoring method may be implemented using the asymptotic likelihood function. This leads to simpler iterative procedures and is commonly used in practice. An example follows.

Example 7.14 - Gaussian Moving Average Process

A common WSS random process has the ACF

$$r_{xx}[k] = \begin{cases} 1 + b^2[1] + b^2[2] & k = 0 \\ b[1] + b[1]b[2] & k = 1 \\ b[2] & k = 2 \\ 0 & k \geq 3. \end{cases}$$

The PSD is easily shown to be

$$P_{xx}(f) = |1 + b[1]\exp(-j2\pi f) + b[2]\exp(-j4\pi f)|^2 .$$

The process $x[n]$ is obtained by passing WGN of variance 1 through a filter with system function $\mathcal{B}(z) = 1 + b[1]z^{-1} + b[2]z^{-2}$. It is usually assumed that the filter $\mathcal{B}(z)$ is minimum-phase or that the zeros z_1, z_2 are within the unit circle. This is termed a moving average (MA) process of order 2. In finding the MLE of the MA filter parameters $b[1], b[2]$, we will need to invert the covariance matrix. Instead, we can use (7.60) to obtain an approximate MLE. Noting that the filter is minimum-phase

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \ln P_{xx}(f)\, df = 0$$

for this process (see Problem 7.22), the approximate MLE is found by minimizing

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{I(f)}{|1 + b[1]\exp(-j2\pi f) + b[2]\exp(-j4\pi f)|^2}\, df.$$

However,

$$\mathcal{B}(z) = (1 - z_1 z^{-1})(1 - z_2 z^{-1})$$

with $|z_1| < 1, |z_2| < 1$. The MLE is found by minimizing

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{I(f)}{|1 - z_1\exp(-j2\pi f)|^2 |1 - z_2\exp(-j2\pi f)|^2}\, df$$

over the allowable zero values and then converting to the MA parameters by

$$\begin{aligned} b[1] &= -(z_1 + z_2) \\ b[2] &= z_1 z_2. \end{aligned}$$

For values of z_1 that are complex we would employ the constraint $z_2 = z_1^*$. For real values of z_1 we must ensure that z_2 is also real. These constraints are necessitated by the coefficients of $\mathcal{B}(z)$ being real. Either a grid search or one of the iterative techniques could then be used. ◇

Another example of the asymptotic MLE is given in the next section.

7.10 Signal Processing Examples

In Section 3.11 we discussed four signal processing estimation examples. The CRLB was determined for each one. By the asymptotic properties of the MLE the asymptotic PDF of the MLE is given by $\mathcal{N}(\boldsymbol{\theta}, \mathbf{I}^{-1}(\boldsymbol{\theta}))$, where $\mathbf{I}^{-1}(\boldsymbol{\theta})$ is given in Section 3.11. We now determine the explicit form for the MLEs. The reader should consult Section 3.11 for the problem statements and notation.

Example 7.15 - Range Estimation (Example 3.13)

From (3.36) the PDF is seen to be

$$\begin{aligned} p(\mathbf{x}; n_0) &= \prod_{n=0}^{n_0-1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2}x^2[n]\right] \\ &\quad \cdot \prod_{n=n_0}^{n_0+M-1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2}(x[n]-s[n-n_0])^2\right] \\ &\quad \cdot \prod_{n=n_0+M}^{N-1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2}x^2[n]\right]. \end{aligned}$$

In this case the continuous parameter τ_0 has been discretized as $n_0 = \tau_0/\Delta$. The likelihood function simplifies to

$$\begin{aligned} p(\mathbf{x}; n_0) &= \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1} x^2[n]\right] \\ &\quad \cdot \prod_{n=n_0}^{n_0+M-1} \exp\left[-\frac{1}{2\sigma^2}(-2x[n]s[n-n_0]+s^2[n-n_0])\right]. \end{aligned}$$

The MLE of n_0 is found by maximizing

$$\exp\left[-\frac{1}{2\sigma^2}\sum_{n=n_0}^{n_0+M-1}(-2x[n]s[n-n_0]+s^2[n-n_0])\right]$$

or, equivalently, by minimizing

$$\sum_{n=n_0}^{n_0+M-1}(-2x[n]s[n-n_0]+s^2[n-n_0]).$$

But $\sum_{n=n_0}^{n_0+M-1} s^2[n-n_0] = \sum_{n=0}^{N-1} s^2[n]$ and is not a function of n_0. Hence, the MLE of n_0 is found by maximizing

$$\sum_{n=n_0}^{n_0+M-1} x[n]s[n-n_0]. \tag{7.62}$$

By the invariance principle, since $R = c\tau_0/2 = cn_0\Delta/2$, the MLE of range is $\hat{R} = (c\Delta/2)\hat{n}_0$. Note that the MLE of the delay n_0 is found by *correlating* the data with all possible received signals and then choosing the maximum. ◇

Example 7.16 - Sinusoidal Parameter Estimation (Example 3.14)

The PDF is given as

$$p(\mathbf{x};\boldsymbol{\theta}) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n] - A\cos(2\pi f_0 n + \phi))^2\right]$$

where $A > 0$ and $0 < f_0 < 1/2$. The MLE of amplitude A, frequency f_0, and phase ϕ is found by minimizing

$$J(A, f_0, \phi) = \sum_{n=0}^{N-1}(x[n] - A\cos(2\pi f_0 n + \phi))^2.$$

We first expand the cosine to yield

$$J(A, f_0, \phi) = \sum_{n=0}^{N-1}(x[n] - A\cos\phi\cos 2\pi f_0 n + A\sin\phi\sin 2\pi f_0 n)^2.$$

Although J is nonquadratic in A and ϕ, we may transform it to a quadratic function by letting

$$\begin{aligned} \alpha_1 &= A\cos\phi \\ \alpha_2 &= -A\sin\phi \end{aligned}$$

which is a one-to-one transformation. The inverse transformation is given by

$$\begin{aligned} A &= \sqrt{\alpha_1^2 + \alpha_2^2} \\ \phi &= \arctan\left(\frac{-\alpha_2}{\alpha_1}\right). \end{aligned} \tag{7.63}$$

Also, let

$$\begin{aligned} \mathbf{c} &= [1\ \cos 2\pi f_0 \ldots \cos 2\pi f_0(N-1)]^T \\ \mathbf{s} &= [0\ \sin 2\pi f_0 \ldots \sin 2\pi f_0(N-1)]^T. \end{aligned}$$

Then, we have

$$\begin{aligned} J'(\alpha_1, \alpha_2, f_0) &= (\mathbf{x} - \alpha_1\mathbf{c} - \alpha_2\mathbf{s})^T(\mathbf{x} - \alpha_1\mathbf{c} - \alpha_2\mathbf{s}) \\ &= (\mathbf{x} - \mathbf{H}\boldsymbol{\alpha})^T(\mathbf{x} - \mathbf{H}\boldsymbol{\alpha}) \end{aligned}$$

where $\boldsymbol{\alpha} = [\alpha_1\ \alpha_2]^T$ and $\mathbf{H} = [\mathbf{c}\ \mathbf{s}]$. Hence, the function to be minimized over $\boldsymbol{\alpha}$ is *exactly that encountered* in the linear model in (7.42) with $\mathbf{C} = \mathbf{I}$. The minimizing solution is, from (7.43),

$$\hat{\boldsymbol{\alpha}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x} \tag{7.64}$$

so that

$$\begin{aligned} J'(\hat{\alpha}_1, \hat{\alpha}_2, f_0) &= (\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\alpha}})^T(\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\alpha}}) \\ &= (\mathbf{x} - \mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x})^T(\mathbf{x} - \mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}) \\ &= \mathbf{x}^T(\mathbf{I} - \mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T)\mathbf{x} \end{aligned}$$

since $\mathbf{I} - \mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T$ is an idempotent matrix. (A matrix is idempotent if $\mathbf{A}^2 = \mathbf{A}$.) Hence, to find $\hat{f}_0$ we need to minimize J' over f_0 or, equivalently, maximize

$$\mathbf{x}^T\mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}.$$

Using the definition of $\mathbf{H}$, we have the MLE for frequency as the value that maximizes

$$\begin{bmatrix} \mathbf{c}^T\mathbf{x} \\ \mathbf{s}^T\mathbf{x} \end{bmatrix}^T \begin{bmatrix} \mathbf{c}^T\mathbf{c} & \mathbf{c}^T\mathbf{s} \\ \mathbf{s}^T\mathbf{c} & \mathbf{s}^T\mathbf{s} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{c}^T\mathbf{x} \\ \mathbf{s}^T\mathbf{x} \end{bmatrix}. \tag{7.65}$$

Once $\hat{f}_0$ is found from the above expression, $\hat{\boldsymbol{\alpha}}$ can be found from (7.64), and then $\hat{A}, \hat{\phi}$ from (7.63). An *approximate* MLE can be obtained if f_0 is not near 0 or 1/2 (see also Problem 7.24) since in this case

$$\begin{aligned} \frac{1}{N}\mathbf{c}^T\mathbf{s} &= \frac{1}{N}\sum_{n=0}^{N-1} \cos 2\pi f_0 n \sin 2\pi f_0 n \\ &= \frac{1}{2N}\sum_{n=0}^{N-1} \sin 4\pi f_0 n \\ &\approx 0 \end{aligned}$$

and, similarly, $\mathbf{c}^T\mathbf{c}/N \approx 1/2$ and $\mathbf{s}^T\mathbf{s}/N \approx 1/2$ (see Problem 3.7). Then, (7.65) becomes approximately

$$\begin{bmatrix} \mathbf{c}^T\mathbf{x} \\ \mathbf{s}^T\mathbf{x} \end{bmatrix}^T \begin{bmatrix} \frac{N}{2} & 0 \\ 0 & \frac{N}{2} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{c}^T\mathbf{x} \\ \mathbf{s}^T\mathbf{x} \end{bmatrix}$$

$$\begin{aligned} &= \frac{2}{N}\left[(\mathbf{c}^T\mathbf{x})^2 + (\mathbf{s}^T\mathbf{x})^2\right] \\ &= \frac{2}{N}\left[\left(\sum_{n=0}^{N-1} x[n]\cos 2\pi f_0 n\right)^2 + \left(\sum_{n=0}^{N-1} x[n]\sin 2\pi f_0 n\right)^2\right] \\ &= \frac{2}{N}\left|\sum_{n=0}^{N-1} x[n]\exp(-j2\pi f_0 n)\right|^2 \end{aligned}$$

or the MLE of frequency is obtained by maximizing the periodogram

$$I(f) = \frac{1}{N}\left|\sum_{n=0}^{N-1} x[n]\exp(-j2\pi fn)\right|^2 \tag{7.66}$$

over f. Then, from (7.64)

$$\begin{aligned}\hat{\boldsymbol{\alpha}} &\approx \frac{2}{N}\begin{bmatrix}\hat{\mathbf{c}}^T\mathbf{x}\\ \hat{\mathbf{s}}^T\mathbf{x}\end{bmatrix}\\ &= \begin{bmatrix}\frac{2}{N}\sum_{n=0}^{N-1} x[n]\cos 2\pi\hat{f}_0 n\\ \frac{2}{N}\sum_{n=0}^{N-1} x[n]\sin 2\pi\hat{f}_0 n\end{bmatrix}\end{aligned}$$

and thus finally we have

$$\begin{aligned}\hat{A} &= \sqrt{\hat{\alpha}_1{}^2+\hat{\alpha}_2{}^2} = \frac{2}{N}\left|\sum_{n=0}^{N-1} x[n]\exp(-j2\pi\hat{f}_0 n)\right|\\ \hat{\phi} &= \arctan\frac{-\sum_{n=0}^{N-1} x[n]\sin 2\pi\hat{f}_0 n}{\sum_{n=0}^{N-1} x[n]\cos 2\pi\hat{f}_0 n}.\end{aligned}$$

◇

Example 7.17 - Bearing Estimation (Example 3.15)

The bearing β of a sinusoidal signal impinging on a line array is related to the spatial frequency f_s as

$$f_s = \frac{F_0 d}{c}\cos\beta.$$

Assuming the amplitude, phase, and spatial frequency are to be estimated, we have the sinusoidal parameter estimation problem just described. It is assumed, however, that the temporal frequency F_0 is known, as well as d and c. Once f_s is estimated using an MLE, the MLE of bearing follows from the invariance property as

$$\hat{\beta} = \arccos\left(\frac{\hat{f}_s c}{F_0 d}\right).$$

To find $\hat{f}_s$ it is common practice to use the approximate MLE obtained from the peak of the periodogram (7.66) or

$$I(f_s) = \frac{1}{M}\left|\sum_{n=0}^{M-1} x[n]\exp(-j2\pi f_s n)\right|^2$$

for M sensors. Here, $x[n]$ is the sample obtained from sensor n. Alternatively, the approximate MLE of bearing can be found directly by maximizing

$$I'(\beta) = \frac{1}{M}\left|\sum_{n=0}^{M-1} x[n]\exp\left(-j2\pi F_0 \frac{d}{c} n\cos\beta\right)\right|^2$$

over β. ◇

Example 7.18 - Autoregressive Parameter Estimation (Example 3.16)

To find the MLE we will use the asymptotic form of the log-likelihood given by (7.60). Then, since the PSD is

$$P_{xx}(f) = \frac{\sigma_u^2}{|A(f)|^2}$$

we have

$$\ln p(\mathbf{x};\mathbf{a},\sigma_u^2) = -\frac{N}{2}\ln 2\pi - \frac{N}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}}\left[\ln\frac{\sigma_u^2}{|A(f)|^2} + \frac{I(f)}{\frac{\sigma_u^2}{|A(f)|^2}}\right] df.$$

As shown in Problem 7.22, since $\mathcal{A}(z)$ is minimum-phase (required for the stability of $1/\mathcal{A}(z)$), then

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \ln|A(f)|^2 df = 0$$

and therefore

$$\ln p(\mathbf{x};\mathbf{a},\sigma_u^2) = -\frac{N}{2}\ln 2\pi - \frac{N}{2}\ln\sigma_u^2 - \frac{N}{2\sigma_u^2}\int_{-\frac{1}{2}}^{\frac{1}{2}} |A(f)|^2 I(f)\, df. \tag{7.67}$$

Differentiating with respect to σ_u^2 and setting the result equal to zero produces

$$-\frac{N}{2\sigma_u^2} + \frac{N}{2\sigma_u^4}\int_{-\frac{1}{2}}^{\frac{1}{2}} |A(f)|^2 I(f)\, df = 0$$

and thus

$$\hat{\sigma_u^2} = \int_{-\frac{1}{2}}^{\frac{1}{2}} |A(f)|^2 I(f)\, df.$$

Substituting this into (7.67) results in

$$\ln p(\mathbf{x};\mathbf{a},\hat{\sigma_u^2}) = -\frac{N}{2}\ln 2\pi - \frac{N}{2}\ln\hat{\sigma_u^2} - \frac{N}{2}.$$

To find $\hat{\mathbf{a}}$ we must minimize $\hat{\sigma_u^2}$ or

$$J(\mathbf{a}) = \int_{-\frac{1}{2}}^{\frac{1}{2}} |A(f)|^2 I(f)\,df.$$

Note that this function is quadratic in $\mathbf{a}$, resulting in the global minimum upon differentiation. For $k = 1, 2, \ldots, p$ we have

$$\begin{aligned}\frac{\partial J(\mathbf{a})}{\partial a[k]} &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \left[A(f)\frac{\partial A^*(f)}{\partial a[k]} + \frac{\partial A(f)}{\partial a[k]}A^*(f)\right] I(f)\,df \\ &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \left[A(f)\exp(j2\pi fk) + A^*(f)\exp(-j2\pi fk)\right] I(f)\,df.\end{aligned}$$

Since $A(-f) = A^*(f)$ and $I(-f) = I(f)$, we can rewrite this as

$$\frac{\partial J(\mathbf{a})}{\partial a[k]} = 2\int_{-\frac{1}{2}}^{\frac{1}{2}} A(f)I(f)\exp(j2\pi fk)\,df.$$

Setting it equal to zero produces

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \left(1 + \sum_{l=1}^{p} a[l]\exp(-j2\pi fl)\right) I(f)\exp(j2\pi fk)\,df = 0 \qquad k = 1, 2, \ldots, p \qquad (7.68)$$

or

$$\sum_{l=1}^{p} a[l]\int_{-\frac{1}{2}}^{\frac{1}{2}} I(f)\exp[j2\pi f(k-l)]\,df = -\int_{-\frac{1}{2}}^{\frac{1}{2}} I(f)\exp[j2\pi fk]\,df.$$

But $\int_{-\frac{1}{2}}^{\frac{1}{2}} I(f)\exp[j2\pi fk]df$ is just the inverse Fourier transform of the periodogram evaluated at k. This can be shown to be the estimated ACF (see Problem 7.25):

$$\hat{r}_{xx}[k] = \begin{cases} \dfrac{1}{N}\displaystyle\sum_{n=0}^{N-1-|k|} x[n]x[n+|k|] & |k| \le N-1 \\ 0 & |k| \ge N. \end{cases}$$

The set of equations to be solved for the approximate MLE of the AR filter parameters $\mathbf{a}$ becomes

$$\sum_{l=1}^{p} \hat{a}[l]\hat{r}_{xx}[k-l] = -\hat{r}_{xx}[k] \qquad k = 1, 2, \ldots, p$$

or in matrix form

$$\begin{bmatrix} \hat{r}_{xx}[0] & \hat{r}_{xx}[1] & \dots & \hat{r}_{xx}[p-1] \\ \hat{r}_{xx}[1] & \hat{r}_{xx}[0] & \dots & \hat{r}_{xx}[p-2] \\ \vdots & \vdots & \ddots & \vdots \\ \hat{r}_{xx}[p-1] & \hat{r}_{xx}[p-2] & \dots & \hat{r}_{xx}[0] \end{bmatrix} \begin{bmatrix} \hat{a}[1] \\ \hat{a}[2] \\ \vdots \\ \hat{a}[p] \end{bmatrix} = - \begin{bmatrix} \hat{r}_{xx}[1] \\ \hat{r}_{xx}[2] \\ \vdots \\ \hat{r}_{xx}[p] \end{bmatrix}. \tag{7.69}$$

These are the so-called *estimated Yule-Walker equations* and this is the *autocorrelation method of linear prediction.* Note the special form of the matrix and the right-hand vector, thereby allowing a recursive solution known as the Levinson recursion [Kay 1988]. To complete the discussion we determine an explicit form for the MLE of σ_u^2. From (7.68) we note that

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \hat{A}(f) I(f) \exp(j2\pi fk)\, df = 0 \qquad k = 1, 2, \dots, p.$$

Since

$$\begin{aligned} \hat{\sigma_u^2} &= \int_{-\frac{1}{2}}^{\frac{1}{2}} |\hat{A}(f)|^2 I(f)\, df \\ &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \hat{A}(f) I(f) \hat{A}^*(f)\, df \\ &= \sum_{k=0}^{p} \hat{a}[k] \int_{-\frac{1}{2}}^{\frac{1}{2}} \hat{A}(f) I(f) \exp(j2\pi fk)\, df \end{aligned}$$

we note that all terms in the sum are zero except for $k = 0$ due to (7.69). It follows that (we let $\hat{a}[0] = 1$)

$$\begin{aligned} \hat{\sigma_u^2} &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \hat{A}(f) I(f)\, df \\ &= \sum_{k=0}^{p} \hat{a}[k] \int_{-\frac{1}{2}}^{\frac{1}{2}} I(f) \exp(-j2\pi fk)\, df \\ &= \sum_{k=0}^{p} \hat{a}[k] \hat{r}_{xx}[-k] \\ &= \sum_{k=0}^{p} \hat{a}[k] \hat{r}_{xx}[k]. \end{aligned}$$

Finally, the MLE is given by

$$\hat{\sigma_u^2} = \hat{r}_{xx}[0] + \sum_{k=1}^{p} \hat{a}[k] \hat{r}_{xx}[k]. \tag{7.70}$$

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Problems

7.1 Show that the CRLB for $\hat{A}$ in Example 7.1 is given by (7.2).

7.2 Consider the sample mean estimator for the problem in Example 7.1. Find the variance and compare it to the CRLB. Does the sample mean estimator attain the CRLB for finite N? How about if $N \to \infty$? Is the MLE or the sample mean a better estimator?

7.3 We observe N IID samples from the PDFs:

a. Gaussian

$$p(x;\mu) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}(x-\mu)^2\right].$$

b. Exponential

$$p(x;\lambda) = \begin{cases} \lambda \exp(-\lambda x) & x > 0 \\ 0 & x < 0. \end{cases}$$

In each case find the MLE of the unknown parameter and be sure to verify that it indeed maximizes the likelihood function. Do the estimators make sense?

7.4 Asymptotic results can sometimes be misleading and so should be carefully applied. As an example, for two unbiased estimators of θ the variances are given by

$$\begin{aligned} \text{var}(\hat{\theta}_1) &= \frac{2}{N} \\ \text{var}(\hat{\theta}_2) &= \frac{1}{N} + \frac{100}{N^2}. \end{aligned}$$

Plot the variances versus N to determine the better estimator.

7.5 A formal definition of the consistency of an estimator is given as follows. An estimator $\hat{\theta}$ is consistent if, given any $\epsilon > 0$,

$$\lim_{N \to \infty} \Pr\{|\hat{\theta} - \theta| > \epsilon\} = 0.$$

Prove that the sample mean is a consistent estimator for the problem of estimating a DC level A in white Gaussian noise of known variance. Hint: Use Chebychev's inequality.

7.6 Another consistency result asserts that if $\alpha = g(\theta)$ for g, a continuous function, and $\hat{\theta}$ is consistent for θ, then $\hat{\alpha} = g(\hat{\theta})$ is consistent for α. Using the statistical linearization argument and the formal definition of consistency, show why this is true. Hint: Linearize g about the true value of θ.

7.7 Consider N IID observations from the exponential family of PDFs

$$p(x;\theta) = \exp\left[A(\theta)B(x) + C(x) + D(\theta)\right]$$

where A, B, C, and D are functions of their respective arguments. Find an equation to be solved for the MLE. Now apply your results to the PDFs in Problem 7.3.

7.8 If we observe N IID samples from a Bernoulli experiment (coin toss) with the probabilities

$$\begin{aligned} \Pr\{x[n] = 1\} &= p \\ \Pr\{x[n] = 0\} &= 1 - p \end{aligned}$$

find the MLE of p.

7.9 For N IID observations from a $\mathcal{U}[0, \theta]$ PDF find the MLE of θ.

7.10 If the data set

$$x[n] = As[n] + w[n] \qquad n = 0, 1, \ldots, N-1$$

is observed, where $s[n]$ is known and $w[n]$ is WGN with known variance σ^2, find the MLE of A. Determine the PDF of the MLE and whether or not the asymptotic PDF holds.

7.11 Find an equation to be solved for the MLE of the correlation coefficient ρ in Problem 3.15. Can you find an approximate solution if $N \to \infty$? Hint: Let $\sum_{n=0}^{N-1} x_1^2[n]/N \to 1$ and $\sum_{n=0}^{N-1} x_2^2[n]/N \to 1$ in the equation where $\mathbf{x}[n] = [x_1[n]\, x_2[n]]^T$.

7.12 In this problem we prove that if an efficient estimator exists, the maximum likelihood method will produce it. Assuming a scalar parameter, if an efficient estimator exists, then we have

$$\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} = I(\theta)(\hat{\theta} - \theta).$$

Use this to prove the theorem.

7.13 We observe N IID samples of a DC level in WGN or

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where A is to be estimated and $w[n]$ is WGN with known variance σ^2. Using a Monte Carlo computer simulation, verify that the PDF of the MLE or sample mean is $\mathcal{N}(A, \sigma^2/N)$. Plot the theoretical and computer-generated PDFs for comparison. Use $A = 1$, $\sigma^2 = 0.1$, $N = 50$, and $M = 1000$ realizations. What happens if M is increased to 5000? Hint: Use the computer subroutines given in Appendix 7A.

7.14 Consider the IID data samples $x[n]$ for $n = 0, 1, \ldots, N-1$, where $x[n] \sim \mathcal{N}(0, \sigma^2)$. Then, according to Slutsky's theorem [Bickel and Doksum 1977], if $\bar{x}$ denotes the sample mean and

$$\hat{\sigma^2} = \frac{1}{N}\sum_{n=0}^{N-1}(x[n] - \bar{x})^2$$

denotes the sample variance, then

$$\frac{\bar{x}}{\hat{\sigma}/\sqrt{N}} \overset{a}{\sim} \mathcal{N}(0,1).$$

Although this may be proven analytically, it requires familiarity with convergence of random variables. Instead, implement a Monte Carlo computer simulation to find the PDF of $\bar{x}/(\hat{\sigma}/\sqrt{N})$ for $\sigma^2 = 1$ and $N = 10$, $N = 100$. Compare your results to the theoretical asymptotic PDF. Hint: Use the computer subroutines given in Appendix 7A.

7.15 In this problem we show that the MLE attains its asymptotic PDF when the estimation error is small. Consider Example 7.6 and let

$$\begin{aligned} \epsilon_s &= -\frac{2}{NA}\sum_{n=0}^{N-1} w[n]\sin 2\pi f_0 n \\ \epsilon_c &= \frac{2}{NA}\sum_{n=0}^{N-1} w[n]\cos 2\pi f_0 n \end{aligned}$$

in (7.20). Assume that f_0 is not near 0 or 1/2, so that the same approximations can be made as in Example 3.4. Prove first that ϵ_s and ϵ_c are approximately

uncorrelated and hence independent Gaussian random variables. Next, determine the PDF of ϵ_s, ϵ_c. Then, assuming ϵ_s and ϵ_c to be small, use a truncated Taylor series expansion of $\hat{\phi}$ about the true value ϕ. This will yield

$$\begin{aligned}\hat{\phi} &= g(\epsilon_s, \epsilon_c) \\ &\approx g(0,0) + \left.\frac{\partial g(\epsilon_s, \epsilon_c)}{\partial \epsilon_s}\right|_{\epsilon_s=0,\epsilon_c=0} \epsilon_s + \left.\frac{\partial g(\epsilon_s, \epsilon_c)}{\partial \epsilon_c}\right|_{\epsilon_s=0,\epsilon_c=0} \epsilon_c\end{aligned}$$

where the function g is given in (7.20). Use this expansion to find the PDF of $\hat{\phi}$. Compare the variance with the CRLB in Example 3.4. Note that for ϵ_s, ϵ_c to be small, either $N \to \infty$ and/or $A \to \infty$.

7.16 For Example 7.7 determine the var$(\hat{A})$ as well as the CRLB for the non-Gaussian PDF (termed a *Laplacian* PDF)

$$p(w[0]) = \frac{1}{2}\exp(-|w[0]|).$$

Does the MLE attain the CRLB as $N \to \infty$?

7.17 For N IID observations from a $\mathcal{N}(0, 1/\theta)$ PDF, where $\theta > 0$, find the MLE of θ and its asymptotic PDF.

7.18 Plot the function

$$g(x) = \exp\left[-\frac{1}{2}x^2\right] + 0.1\exp\left[-\frac{1}{2}(x-10)^2\right]$$

over the domain $-3 \leq x \leq 13$. Find the maximum of the function from the graph. Then, use a Newton-Raphson iteration to find the maximum. In doing so use the initial guesses of $x_0 = 0.5$ and $x_0 = 9.5$. What can you say about the importance of the initial guess?

7.19 If

$$x[n] = \cos 2\pi f_0 n + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN with known variance σ^2, show that the MLE of frequency is obtained approximately by maximizing (f_0 not near 0 or 1/2)

$$\sum_{n=0}^{N-1} x[n]\cos 2\pi f_0 n$$

over the interval $0 < f_0 < 1/2$. Next, use a Monte Carlo computer simulation (see Appendix 7A) for $N = 10, f_0 = 0.25, \sigma^2 = 0.01$ and plot the function to be maximized. Apply the Newton-Raphson method for determining the maximum and compare the results to those from a grid search.

7.20 Consider the data set

$$x[n] = s[n] + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $s[n]$ is unknown for $n = 0, 1, \ldots, N-1$ and $w[n]$ is WGN with known variance σ^2. Determine the MLE of $s[n]$ and also its PDF. Do the asymptotic MLE properties hold? That is, is the MLE unbiased, efficient, Gaussian, and consistent?

7.21 For N IID observation from the PDF $\mathcal{N}(A, \sigma^2)$, where A and σ^2 are both unknown, find the MLE of the SNR $\alpha = A^2/\sigma^2$.

7.22 Prove that if

$$\mathcal{A}(z) = 1 + \sum_{k=1}^{p} a[k] z^{-k}$$

is a minimum-phase polynomial (all roots inside the unit circle of the z plane), then

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \ln |A(f)|^2 \, df = 0.$$

To do so first show that

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \ln |A(f)|^2 \, df = 2 \operatorname{Re} \left\{ \frac{1}{2\pi j} \oint \ln \mathcal{A}(z) \frac{dz}{z} \right\}$$

where the contour is the unit circle in the z plane. Next, note that

$$\frac{1}{2\pi j} \oint \ln \mathcal{A}(z) \frac{dz}{z}$$

is the inverse z transform of $\ln \mathcal{A}(z)$ evaluated at $n = 0$. Finally, use the fact that a minimum-phase $\mathcal{A}(z)$ results in an inverse z transform of $\ln \mathcal{A}(z)$ that is causal.

7.23 Find the asymptotic MLE for the total power P_0 of the PSD

$$P_{xx}(f) = P_0 Q(f)$$

where

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} Q(f) \, df = 1.$$

If $Q(f) = 1$ for all f so that the process is WGN, simplify your results. Hint: Use the results from Problem 7.25 for the second part.

7.24 The peak of the periodogram was shown in Example 7.16 to be the MLE for frequency under the condition that f_0 is not near 0 or 1/2. Plot the periodogram for $N = 10$ for the frequencies $f_0 = 0.25$ and $f_0 = 0.05$. Use the noiseless data

$$x[n] = \cos 2\pi f_0 n \qquad n = 0, 1, \ldots, N-1.$$

What happens if this approximation is not valid? Repeat the problem using the exact function

$$\mathbf{x}^T\mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}$$

where $\mathbf{H}$ is defined in Example 7.16.

7.25 Prove that the inverse Fourier transform of the periodogram is

$$\hat{r}_{xx}[k] = \begin{cases} \frac{1}{N}\sum_{n=0}^{N-1-|k|} x[n]x[n+|k|] & |k| \leq N-1 \\ 0 & |k| \geq N. \end{cases}$$

Hint: Note that the periodogram can be written as

$$I(f) = \frac{1}{N}X'(f)X'(f)^*$$

where $X'(f)$ is the Fourier transform of the sequence

$$x'[n] = \begin{cases} x[n] & 0 \leq n \leq N-1 \\ 0 & \text{otherwise.} \end{cases}$$

7.26 An AR(1) PSD given as

$$P_{xx}(f) = \frac{\sigma_u^2}{|1 + a[1]\exp(-j2\pi f)|^2}.$$

If $a[1]$ and σ^2 are unknown, find the MLE of $P_{xx}(f)$ for $f = f_0$ by using (7.69) and (7.70). What is the asymptotic variance of the MLE? Hint: Use (3.30) and (3.44).

Appendix 7A

Monte Carlo Methods

We now describe the computer methods employed to generate realizations of the random variable (see Example 7.1)

$$\hat{A} = -\frac{1}{2} + \sqrt{\frac{1}{N}\sum_{n=0}^{N-1} x^2[n] + \frac{1}{4}}$$

where $x[n] = A + w[n]$ for $w[n]$ WGN with variance $A > 0$. Additionally, how the statistical properties of $\hat{A}$ such as the mean, variance, and PDF are determined is discussed. The description applies more generally in that the methods are valid for any estimator whose performance is to be determined.

The steps are summarized as follows.

Data Generation:

1. Generate N independent $\mathcal{U}[0, 1]$ random variates.
2. Convert these variates to Gaussian random variates using the Box-Mueller transformation to produce $w[n]$.
3. Add A to $w[n]$ to yield $x[n]$ and then compute $\hat{A}$.
4. Repeat the procedure M times to yield the M realizations of $\hat{A}$.

Statistical Properties:

1. Determine the mean using (7.9).
2. Determine the variance using (7.10).
3. Determine the PDF using a histogram.

This approach is implemented in the accompanying Fortran computer program, MONTECARLO, which we now describe.

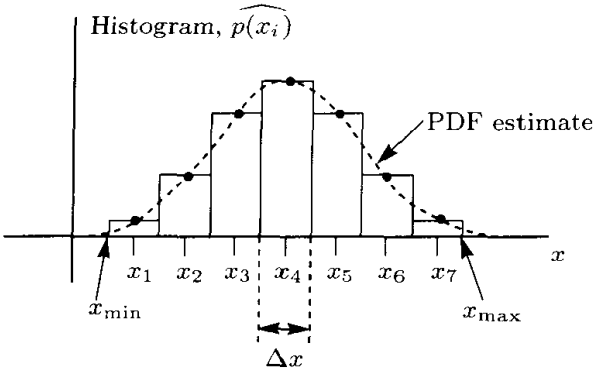


Figure 7A.1 Histogram of computer generated data

In the data generation we first use a standard pseudorandom noise generator which produces independent $\mathcal{U}[0,1]$ random variates. As an example, on a VAX 11/780 the intrinsic function RAN can be used. In MONTECARLO the subroutine RANDOM calls the RAN function N times (for N even). Next, we convert these independent uniformly distributed random variates into independent Gaussian random variates with mean 0 and variance 1 by using the Box-Mueller transformation

$$\begin{aligned} w_1 &= \sqrt{-2\ln u_1}\cos 2\pi u_2 \\ w_2 &= \sqrt{-2\ln u_1}\sin 2\pi u_2 \end{aligned}$$

where u_1, u_2 are independent $\mathcal{U}[0,1]$ random variables and w_1, w_2 are independent $\mathcal{N}(0,1)$ random variables. Since the transformation operates on two random variables at a time, N needs to be even (if odd, just increment by 1 and discard the extra random variate). To convert to $\mathcal{N}(0,\sigma^2)$ random variates simply multiply by σ. This entire procedure is implemented in subroutine WGN. Next, A is added to $w[n]$ to generate one time series realization $x[n]$ for $n = 0, 1, \ldots, N-1$. For this realization of $x[n]$, $\hat{A}$ is computed. We repeat the procedure M times to yield the M realizations of $\hat{A}$.

The mean and variance are determined using (7.9) and (7.10), respectively. The alternative form of the variance estimate

$$\widehat{\text{var}}(\hat{A}) = \frac{1}{M}\sum_{i=1}^{M}\hat{A}_i^2 - \widehat{E(\hat{A})}^2$$

is used by the subroutine STATS. The number of realizations, M, needed to provide an accurate estimate of the mean and variance can be determined by finding the *variance of the estimators* since (7.9) and (7.10) are nothing more than estimators themselves. A simpler procedure is to keep increasing M until the numbers given by (7.9) and (7.10) converge.

Finally, to determine the PDF of $\hat{A}$ we use an estimated PDF called the *histogram*. The histogram estimates the PDF by determining the number of times $\hat{A}$ falls within a specified interval. Then, a division by the total number of realizations to yield the probability, followed by a division by the interval length, produces the PDF estimate. A typical histogram is shown in Figure 7A.1 where the PDF is estimated over the interval

$(x_{\min}, x_{\max})$. Each subinterval $(x_i - \Delta x/2, x_i + \Delta x/2)$ is called a *cell.* The value for the ith cell is found as

$$\widehat{p(x_i)} = \frac{\frac{L_i}{M}}{\Delta x}$$

where L_i is the number of realizations of x_i that lie within the ith cell. Hence, L_i/M estimates the probability that x_i will fall in the ith cell, and by dividing by Δx, we obtain the estimated PDF. If we associate $\widehat{p(x_i)}$ with the PDF at the cell center, then interpolating these points as shown in Figure 7A.1 produces an estimate of the continuous PDF. It is apparent from this discussion that for good estimates of the PDF we would like the cell width to be small. This is because we are estimating not the PDF $p(x)$ but

$$\frac{1}{\Delta x} \int_{x_i - \frac{\Delta x}{2}}^{x_i + \frac{\Delta x}{2}} p(x)\, dx$$

or the average PDF over the cell. Unfortunately, as the cell width becomes smaller (more cells), the probability of a realization falling into the cell becomes smaller. This yields highly variable PDF estimates. As before, a good strategy keeps increasing M until the estimated PDF appears to converge. The histogram approach is implemented in the subroutine HISTOG. A further discussion of PDF estimation can be found in [Bendat and Piersol 1971].

Fortran Program MONTECARLO

```
C  MONTECARLO
C  This program determines the asymptotic properties
C  of the MLE for a N(A,A) PDF (see Examples 7.1-7.3, 7.5).
C
C  The array dimensions are given as variable.  Replace them
C  with numerical values.  To use this program you will need
C  a plotting subroutine to replace PLOT and a random number
C  generator to replace the intrinsic function RAN in the
C  subroutine RANDOM.
C
      DIMENSION X(N),W(N),AHAT(M),PDF(NCELLS),HIST(NCELLS)
     * ,XX(NCELLS)
      PI=4.*ATAN(1.)
C  Input the value of A, the number of data points N, and the
C  number of realizations M.
      WRITE(6,10)
10    FORMAT(' INPUT A, N, AND M')
      READ(5,*)A,N,M
C  Generate M realizations of the estimate of A.
      DO 40 K=1,M
C  Generate the noise samples and add A to each one.
```

```
      CALL WGN(N,A,W)
      DO 20 I=1,N
20    X(I)=A+W(I)
C  Generate the estimate of A.
      XSQ=0.
      DO 30 I=1,N
30    XSQ=XSQ+X(I)*X(I)/N
      AHAT(K)=-0.5+SQRT(XSQ+0.25)
40    CONTINUE
C  Compute the mean and variance of the estimates of A.
      CALL STATS(AHAT,M,AMEAN,AVAR)
C  Normalize the variance by N.
      AVAR=N*AVAR
      WRITE(6,*)AMEAN,AVAR
C  Input the interval (XMIN,XMAX) and the number of cells
C  for the histogram.
      WRITE(6,50)
50    FORMAT(' INPUT XMIN,XMAX,NCELLS FOR HISTOGRAM')
      READ(5,*)XMIN,XMAX,NCELLS
C  Compute the histogram.
      CALL HISTOG(AHAT,M,XMIN,XMAX,NCELLS,XX,HIST)
C  Compute the asymptotic variance.
      SIG2=A*A/(N*(A+0.5))
C  Compute the asymptotic Gaussian PDF for comparison to
C  the histogram.
      DO 60 I=1,NCELLS
      ARG=(XX(I)-A)*(XX(I)-A)/SIG2
      IF(ARG.GT.50.)PDF(I)=0.
      IF(ARG.LT.50.)PDF(I)=(1./SQRT(2.*PI*SIG2))*EXP(-0.5*ARG)
60    CONTINUE
C  Compare graphically the asymptotic PDF to the histogram.
      CALL PLOT(XX,HIST,PDF)
      STOP
      END
      SUBROUTINE STATS(X,N,XMEAN,XVAR)
C  This program computes the sample mean and variance
C  of a data set.
C
C  Input parameters:
C
C    X      - Array of dimension Nx1 containing data
C    N      - Number of data set
C
C  Output parameters:
C
```

```
C     XMEAN - Sample mean
C     XVAR  - Sample variance
C
      DIMENSION X(1)
      XMEAN=0.
      XSQ=0.
      DO 10 I=1,N
      XMEAN=XMEAN+X(I)/N
10    XSQ=XSQ+X(I)*X(I)/N
      XVAR=XSQ-XMEAN*XMEAN
      RETURN
      END
      SUBROUTINE HISTOG(X,N,XMIN,XMAX,NCELLS,XX,HIST)
C  This subroutine computes a histogram of a set of data.
C
C  Input parameters:
C
C     X      -  Input data array of dimension Nx1
C     N      -  Number of data samples
C     XMIN   -  Minimum value for histogram horizontal axis
C     XMAX   -  Maximum value for histogram horizontal axis
C     NCELLS -  Number of cells desired
C
C  Output parameters:
C
C     XX     -  Array of dimension NCELLSx1 containing
C               cell centers
C     HIST   -  Histogram values for each cell where HIST(I)
C               is the value for XX(I), I=1,2,..,NCELLS
C
      DIMENSION X(1),XX(1),HIST(1)
      CELLWID=(XMAX-XMIN)/FLOAT(NCELLS)
      DO 10 I=1,NCELLS
10    XX(I)=XMIN+CELLWID/2.+(I-1.)*CELLWID
      DO 30 K=1,NCELLS
      HIST(K)=0.
      CELLMIN=XX(K)-CELLWID/2.
      CELLMAX=XX(K)+CELLWID/2.
      DO 20 I=1,N
20    IF(X(I).GT.CELLMIN.AND.X(I).LT.CELLMAX)HIST(K)=
     * HIST(K)+1./(N*CELLWID)
30    CONTINUE
      RETURN
      END
      SUBROUTINE PLOT(X,Y1,Y2)
```

```
      DIMENSION X(1),Y1(1),Y2(1)
C  Replace this subroutine with any standard Fortran-compatible
C  plotting routine.
      RETURN
      END
      SUBROUTINE WGN(N,VAR,W)
C  This subroutine generates samples of zero mean white
C  Gaussian noise.
C
C  Input parameters:
C
C    N   - Number of noise samples desired
C    VAR - Variance of noise desired
C
C  Output parameters:
C
C    W   - Array of dimension Nx1 containing noise samples
C
      DIMENSION W(1)
      PI=4.*ATAN(1.)
C   Add 1 to desired number of samples if N is odd.
      N1=N
      IF(MOD(N,2).NE.0)N1=N+1
C   Generate N1 independent and uniformly distributed random
C   variates on [0,1].
      CALL RANDOM(N1,VAR,W)
      L=N1/2
C   Convert uniformly distributed random variates to Gaussian
C   ones using a Box-Mueller transformation.
      DO 10 I=1,L
      U1=W(2*I-1)
      U2=W(2*I)
      TEMP=SQRT(-2.*ALOG(U1))
      W(2*I-1)=TEMP*COS(2.*PI*U2)*SQRT(VAR)
10    W(2*I)=TEMP*SIN(2.*PI*U2)*SQRT(VAR)
      RETURN
      END
      SUBROUTINE RANDOM(N1,VAR,W)
      DIMENSION W(1)
      DO 10 I=1,N1
C   For machines other than DEC VAX 11/780 replace RAN(ISEED)
C   with a random number generator.
10    W(I)=RAN(11111)
      RETURN
      END
```

Appendix 7B

Asymptotic PDF of MLE for a Scalar Parameter

We now give an outline of the proof of Theorem 7.1. A rigorous proof of consistency can be found in [Dudewicz 1976] and of the asymptotic Gaussian property in [Rao 1973]. To simplify the discussion the observations are assumed to be IID. We further assume the following regularity conditions.

1. The first-order and second-order derivatives of the log-likelihood function are well defined.

2.

$$E\left[\frac{\partial \ln p(x[n];\theta)}{\partial \theta}\right] = 0.$$

We first show that the MLE is consistent. To do so we will need the following inequality (see [Dudewicz 1976] for proof), which is related to the Kullback-Leibler information.

$$\int \ln\left[\frac{p(x[n];\theta_1)}{p(x[n];\theta_2)}\right] p(x[n];\theta_1)\, dx[n] \geq 0 \tag{7B.1}$$

with equality if and only if $\theta_1 = \theta_2$. Now, in maximizing the log-likelihood function, we are equivalently maximizing

$$\begin{aligned}\frac{1}{N}\ln p(\mathbf{x};\theta) &= \frac{1}{N}\ln \prod_{n=0}^{N-1} p(x[n];\theta) \\ &= \frac{1}{N}\sum_{n=0}^{N-1} \ln p(x[n];\theta).\end{aligned}$$

But as $N \to \infty$, this converges to the expected value by the law of large numbers. Hence, if θ_0 denotes the true value of θ, we have

$$\frac{1}{N}\sum_{n=0}^{N-1} \ln p(x[n];\theta) \to \int \ln p(x[n];\theta) p(x[n];\theta_0)\, dx[n]. \tag{7B.2}$$

However, from (7B.1)

$$\int \ln\left[p(x[n];\theta_1)\right] p(x[n];\theta_1)\,dx[n] \geq \int \ln\left[p(x[n];\theta_2)\right] p(x[n];\theta_1)\,dx[n].$$

The right-hand side of (7B.2) is therefore maximized for $\theta = \theta_0$. By a suitable continuity argument, the left-hand side of (7B.2) or the normalized log-likelihood function must also be maximized for $\theta = \theta_0$ or as $N \to \infty$, the MLE is $\hat{\theta} = \theta_0$. Thus, the MLE is consistent.

To derive the asymptotic PDF of the MLE we first use a Taylor expansion about the true value of θ, which is θ_0. Then, by the mean value theorem

$$\left.\frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta}\right|_{\theta=\hat{\theta}} = \left.\frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta}\right|_{\theta=\theta_0} + \left.\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2}\right|_{\theta=\tilde{\theta}} (\hat{\theta}-\theta_0)$$

where $\theta_0 < \tilde{\theta} < \hat{\theta}$. But

$$\left.\frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta}\right|_{\theta=\hat{\theta}} = 0$$

by the definition of the MLE, so that

$$0 = \left.\frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta}\right|_{\theta=\theta_0} + \left.\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2}\right|_{\theta=\tilde{\theta}} (\hat{\theta}-\theta_0). \tag{7B.3}$$

Now consider $\sqrt{N}(\hat{\theta}-\theta_0)$, so that (7B.3) becomes

$$\sqrt{N}(\hat{\theta}-\theta_0) = \frac{\left.\frac{1}{\sqrt{N}}\frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta}\right|_{\theta=\theta_0}}{\left.-\frac{1}{N}\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2}\right|_{\theta=\tilde{\theta}}}. \tag{7B.4}$$

Due to the IID assumption

$$\left.\frac{1}{N}\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2}\right|_{\theta=\tilde{\theta}} = \left.\frac{1}{N}\sum_{n=0}^{N-1}\frac{\partial^2 \ln p(x[n];\theta)}{\partial\theta^2}\right|_{\theta=\tilde{\theta}}$$

Since $\theta_0 < \tilde{\theta} < \hat{\theta}$, we must also have $\tilde{\theta} \to \theta_0$ due to the consistency of the MLE. Hence,

$$\begin{aligned}\left.\frac{1}{N}\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2}\right|_{\theta=\tilde{\theta}} &\to \left.\frac{1}{N}\sum_{n=0}^{N-1}\frac{\partial^2 \ln p(x[n];\theta)}{\partial\theta^2}\right|_{\theta=\theta_0} \\ &\to E\left[\left.\frac{\partial^2 \ln p(x[n];\theta)}{\partial\theta^2}\right|_{\theta=\theta_0}\right] \\ &= -i(\theta_0)\end{aligned}$$

where the last convergence is due to the law of large numbers. Also, the numerator term is

$$\frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} \left. \frac{\partial \ln p(x[n];\theta)}{\partial \theta} \right|_{\theta=\theta_0}.$$

Now

$$\xi_n = \left. \frac{\partial \ln p(x[n];\theta)}{\partial \theta} \right|_{\theta=\theta_0}$$

is a random variable, being a function of $x[n]$. Additionally, since the $x[n]$'s are IID, so are the ξ_n's. By the central limit theorem the numerator term in (7B.4) has a PDF that converges to a Gaussian with mean

$$E\left[\frac{1}{\sqrt{N}} \sum_{n-0}^{N-1} \left. \frac{\partial \ln p(x[n];\theta)}{\partial \theta} \right|_{\theta=\theta_0} \right] = 0$$

and variance

$$\begin{aligned} E\left[\left(\frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} \left. \frac{\partial \ln p(x[n];\theta)}{\partial \theta} \right|_{\theta=\theta_0} \right)^2 \right] &= \frac{1}{N} \sum_{n=0}^{N-1} E\left[\left(\frac{\partial \ln p(x[n];\theta)}{\partial \theta} \right)^2 \Bigg|_{\theta=\theta_0} \right] \\ &= i(\theta_0) \end{aligned}$$

due to the independence of the random variables. We now resort to Slutsky's theorem [Bickel and Doksum 1977], which says that if the sequence of random variables x_n has the asymptotic PDF of the random variable x and the sequence of random variables y_n converges to a constant c, then x_n/y_n has the same asymptotic PDF as the random variable x/c. In our case,

$$\begin{aligned} x &\sim \mathcal{N}(0, i(\theta_0)) \\ y_n &\rightarrow c = i(\theta_0) \end{aligned}$$

so that (7B.4) becomes

$$\sqrt{N}(\hat{\theta} - \theta_0) \overset{a}{\sim} \mathcal{N}(0, i^{-1}(\theta_0))$$

or, equivalently,

$$\hat{\theta} \overset{a}{\sim} \mathcal{N}\left(\theta_0, \frac{1}{N i(\theta_0)}\right)$$

or finally

$$\hat{\theta} \overset{a}{\sim} \mathcal{N}(\theta_0, I^{-1}(\theta_0)).$$

Appendix 7C

Derivation of Conditional Log-Likelihood for EM Algorithm Example

From (7.51) we have, upon noting the independent data set assumption,

$$\begin{aligned}
\ln p_y(\mathbf{y};\boldsymbol{\theta}) &= \sum_{i=1}^{p} \ln p(\mathbf{y}_i;\theta_i) \\
&= \sum_{i=1}^{p} \ln \left\{ \frac{1}{(2\pi\sigma_i^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma_i^2} \sum_{n=0}^{N-1} (y_i[n] - \cos 2\pi f_i n)^2 \right] \right\} \\
&= c - \sum_{i=1}^{p} \frac{1}{2\sigma_i^2} \sum_{n=0}^{N-1} (y_i[n] - \cos 2\pi f_i n)^2 \\
&= g(\mathbf{y}) + \sum_{i=1}^{p} \frac{1}{\sigma_i^2} \sum_{n=0}^{N-1} (y_i[n] \cos 2\pi f_i n - \frac{1}{2} \cos^2 2\pi f_i n)
\end{aligned}$$

where c is a constant and $g(\mathbf{y})$ does not depend on the frequencies. Making the approximation that $\sum_{n=0}^{N-1} \cos^2 2\pi f_i n \approx N/2$ for f_i not near 0 or 1/2, we have

$$\ln p_y(\mathbf{y};\boldsymbol{\theta}) = h(\mathbf{y}) + \sum_{i=1}^{p} \frac{1}{\sigma_i^2} \sum_{n=0}^{N-1} y_i[n] \cos 2\pi f_i n$$

or letting $\mathbf{c}_i = [1 \ \cos 2\pi f_i \ldots \cos 2\pi f_i (N-1)]^T$, we have

$$\ln p_y(\mathbf{y};\boldsymbol{\theta}) = h(\mathbf{y}) + \sum_{i=1}^{p} \frac{1}{\sigma_i^2} \mathbf{c}_i^T \mathbf{y}_i. \tag{7C.1}$$

Concatenating the $\mathbf{c}_i$'s and $\mathbf{y}_i$'s as

$$\mathbf{c} = \begin{bmatrix} \frac{1}{\sigma_1^2}\mathbf{c}_1 \\ \frac{1}{\sigma_2^2}\mathbf{c}_2 \\ \vdots \\ \frac{1}{\sigma_p^2}\mathbf{c}_p \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_p \end{bmatrix}$$

we have

$$\ln p_y(\mathbf{y};\boldsymbol{\theta}) = h(\mathbf{y}) + \mathbf{c}^T\mathbf{y}.$$

From (7.56) we write the conditional expectation as

$$\begin{aligned} U(\boldsymbol{\theta},\boldsymbol{\theta}_k) &= E[\ln p_y(\mathbf{y};\boldsymbol{\theta})|\mathbf{x};\boldsymbol{\theta}_k] \\ &= E(h(\mathbf{y})|\mathbf{x};\boldsymbol{\theta}_k) + \mathbf{c}^T E(\mathbf{y}|\mathbf{x};\boldsymbol{\theta}_k). \end{aligned} \tag{7C.2}$$

Since we wish to maximize $U(\boldsymbol{\theta},\boldsymbol{\theta}_k)$ with respect to $\boldsymbol{\theta}$, we can omit the expected value of $h(\mathbf{y})$ since it does not depend on $\boldsymbol{\theta}$. Now, $\mathbf{y}$ and $\mathbf{x}$ are jointly Gaussian since, from (7.52),

$$\mathbf{x} = \sum_{i=1}^{p} \mathbf{y}_i = [\mathbf{I}\ \mathbf{I} \ldots \mathbf{I}]\mathbf{y}$$

where $\mathbf{I}$ is the $N \times N$ identity matrix and the transformation matrix is composed of p identity matrices. Using the standard result for conditional expectations of jointly Gaussian random vectors (see Appendix 10A), we have

$$E(\mathbf{y}|\mathbf{x};\boldsymbol{\theta}_k) = E(\mathbf{y}) + \mathbf{C}_{yx}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})).$$

The means are given by

$$\begin{aligned} E(\mathbf{y}) &= \begin{bmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \\ \vdots \\ \mathbf{c}_p \end{bmatrix} \\ E(\mathbf{x}) &= \sum_{i=1}^{p} \mathbf{c}_i \end{aligned}$$

while the covariance matrices are

$$\begin{aligned} \mathbf{C}_{xx} &= \sigma^2\mathbf{I} \\ \mathbf{C}_{yx} &= E\left(\begin{bmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \vdots \\ \mathbf{w}_p \end{bmatrix} \mathbf{w}^T \right) \end{aligned}$$

$$
\begin{aligned}
&= E\left(\begin{bmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \vdots \\ \mathbf{w}_p \end{bmatrix} \left\{ [\mathbf{I}\ \mathbf{I} \ldots \mathbf{I}] \begin{bmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \vdots \\ \mathbf{w}_p \end{bmatrix} \right\}^T \right) \\
&= \begin{bmatrix} \sigma_1^2\mathbf{I} & \mathbf{0} & \ldots & \mathbf{0} \\ \mathbf{0} & \sigma_2^2\mathbf{I} & \ldots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \ldots & \sigma_p^2\mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{I} \\ \mathbf{I} \\ \vdots \\ \mathbf{I} \end{bmatrix} \\
&= \begin{bmatrix} \sigma_1^2\mathbf{I} \\ \sigma_2^2\mathbf{I} \\ \vdots \\ \sigma_p^2\mathbf{I} \end{bmatrix},
\end{aligned}
$$

so that

$$
\begin{aligned}
E(\mathbf{y}|\mathbf{x};\boldsymbol{\theta}_k) &= \begin{bmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \\ \vdots \\ \mathbf{c}_p \end{bmatrix} + \frac{1}{\sigma^2} \begin{bmatrix} \sigma_1^2\mathbf{I} \\ \sigma_2^2\mathbf{I} \\ \vdots \\ \sigma_p^2\mathbf{I} \end{bmatrix} \left(\mathbf{x} - \sum_{i=1}^{p} \mathbf{c}_i \right) \\
&= \begin{bmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \\ \vdots \\ \mathbf{c}_p \end{bmatrix} + \begin{bmatrix} \frac{\sigma_1^2}{\sigma^2}(\mathbf{x} - \sum_{i=1}^{p} \mathbf{c}_i) \\ \frac{\sigma_2^2}{\sigma^2}(\mathbf{x} - \sum_{i=1}^{p} \mathbf{c}_i) \\ \vdots \\ \frac{\sigma_p^2}{\sigma^2}(\mathbf{x} - \sum_{i=1}^{p} \mathbf{c}_i) \end{bmatrix}
\end{aligned}
$$

or

$$
E(\mathbf{y}_i|\mathbf{x};\boldsymbol{\theta}_k) = \mathbf{c}_i + \frac{\sigma_i^2}{\sigma^2}(\mathbf{x} - \sum_{i=1}^{p} \mathbf{c}_i) \qquad i = 1, 2, \ldots, p
$$

where $\mathbf{c}_i$ is computed using $\boldsymbol{\theta}_k$. Note that $E(\mathbf{y}_i|\mathbf{x};\boldsymbol{\theta}_k)$ can be thought of as an estimate of the $y_i[n]$ data set since, letting $\hat{\mathbf{y}}_i = E(\mathbf{y}_i|\mathbf{x};\boldsymbol{\theta}_k)$,

$$
\hat{y}_i[n] = \cos 2\pi f_{i_k} n + \frac{\sigma_i^2}{\sigma^2} \left(x[n] - \sum_{i=1}^{p} \cos 2\pi f_{i_k} n \right). \tag{7C.3}
$$

Using (7C.2) and dropping the term that is independent of $\boldsymbol{\theta}$, we have

$$
U'(\boldsymbol{\theta}, \boldsymbol{\theta}_k) = \sum_{i=1}^{p} \mathbf{c}_i^T \hat{\mathbf{y}}_i
$$

which is maximized over $\boldsymbol{\theta}$ *by maximizing each term in the sum separately* or

$$f_{i_{k+1}} = \arg\max_{f_i} \mathbf{c}_i^T \hat{\mathbf{y}}_i. \tag{7C.4}$$

Finally, since the σ_i^2's are not unique, they can be chosen arbitrarily as long as (see (7.53))

$$\sum_{i=1}^{p} \sigma_i^2 = \sigma^2$$

or, equivalently,

$$\sum_{i=1}^{p} \beta_i = \sum_{i=1}^{p} \frac{\sigma_i^2}{\sigma^2} = 1.$$